

***TM 1-1520-237-CL**

**TECHNICAL MANUAL
OPERATOR'S AND
CREWMEMBER'S
CHECKLIST**

**ARMY MODELS
UH-60A HELICOPTER
UH-60L HELICOPTER
EH-60A HELICOPTER**

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE: Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

***This manual supersedes TM 1-1520-237-CL, dated 1 May 2003, including all changes.**

**HEADQUARTERS
DEPARTMENT OF THE
ARMY**

17 April 2006

CHANGE HEADQUARTERS
NO. 3 DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 20 June 2008

**TECHNICAL MANUAL
OPERATOR'S AND CREWMEMBER'S
CHECKLIST**

**ARMY MODELS
UH-60A HELICOPTER
UH-60L HELICOPTER
EH-60A HELICOPTER**

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-CL, dated 17 April 2006, is changed as follows:

1. Remove and insert pages as indicated below. On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by a hand pointing to the changed area on the illustration or a MAJOR CHANGE symbol.

TM 1-1520-237-CL
C3

Text that flows to the following page is indicated by the current change number.

Remove pages

A and B
iii/(iv Blank)
N-3 and N-4
N-4.1/(N-4.2 Blank)
N-13 through N-18
E-13 and E-14
P-3 and P-4

Insert pages

A and B
iii/(iv Blank)
N-3 and N-4
N-4.1/(N-4.2 Blank)
N-13 through N-18
E-13 and E-14
P-3 and P-4

2. Retain this sheet in front of manual for reference purposes.

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:



JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0815622

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311310,, requirements for TM 1-1520-237-CL.

CHANGE HEADQUARTERS
NO. 2 DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 15 December 2007

**TECHNICAL MANUAL
OPERATOR'S AND CREWMEMBER'S
CHECKLIST**

**ARMY MODELS
UH-60A HELICOPTER
UH-60L HELICOPTER
EH-60A HELICOPTER**

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-CL, dated 17 April 2006, is changed as follows:

1. Remove and insert pages as indicated below. On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by a hand pointing to the changed area on the illustration or a MAJOR CHANGE symbol.

TM 1-1520-237-CL
C2

Text that flows to the following page is indicated by the current change number.

Remove pages

A and B
iii/(iv Blank)
N-1 through N-4

N-9 and N-10
N-11/(N-12 Blank)
N-15 through N-18
E-29 and E-30
P-3 and P-4

Insert pages

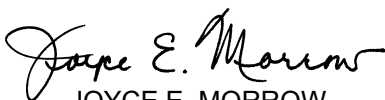
A and B
iii/(iv Blank)
N-1 through N-4
N-4.1/(N-4.2 Blank)
N-9 and N-10
N-11/(N-12 Blank)
N-15 through N-18
E-29 and E-30
P-3 and P-4

2. Retain this sheet in front of manual for reference purposes.

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:



JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army

0734706

Distribution:

To be distributed in accordance with the initial distribution number (IDN) 311310, requirements for TM 1-1520-237-CL.

CHANGE HEADQUARTERS
NO. 1 DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 21 March 2007

**TECHNICAL MANUAL
OPERATOR'S AND CREWMEMBER'S
CHECKLIST**

**ARMY MODELS
UH-60A HELICOPTER
UH-60L HELICOPTER
EH-60A HELICOPTER**

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-CL, dated 17 April 2006, is changed as follows:

1. Remove and insert pages as indicated below. On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by a hand pointing to the changed area on the illustration or a MAJOR CHANGE symbol.

Text that flows to the following page is indicated by the current change number.

Remove pages

A/(B Blank)
N-3 through N-10
N-11/(N-12 Blank)
N-15 through N-18
E-3 through E-6
E-19 and E-20
E-31 through E-34
E-35/(E-36 Blank)
P-3 and P-4
P-9 and P-10
P-19 through P-22

P-27 and P-28
P-29/(P-30 Blank)
P-59 through P-64
P-65/(P-66 Blank)
P-67/(P-68 Blank)
P-69 through P-86

Insert pages

A and B
N-3 through N-10
N-11/(N-12 Blank)
N-15 through N-18
E-3 through E-6
E-19 and E-20
E-31 through E-34
E-35/(E-36 Blank)
P-3 and P-4
P-9 and P-10
P-19 through P-22
P-22.1/(P-22.2 Blank)
P-27 and P-28
P-29 and P-30
P-59 through P-64
P-65/(P-66 Blank)
P-67/(P-68 Blank)
P-69 through P-86

2. Retain these sheets in front of manual for reference purposes.

By Order of the Secretary of the Army:

PPETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is written in a cursive, flowing style.

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0703605

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311310, requirements for TM 1-1520-237-CL.

LIST OF EFFECTIVE PAGES

Insert latest changed pages; dispose of superseded pages in accordance with applicable regulations.

NOTE: On a changed page, the portion of the text affected by the latest change is indicated by a vertical line in the outer margin of the page. Changes to illustrations are indicated by a hand pointing to the changed area on the illustration or a MAJOR CHANGE symbol.

Dates of issue for original and changed pages are:

Original	17 April 2006
Change 1	21 March 2007
Change 2	15 December 2007
Change 3	20 June 2008

Total number of pages in this manual is 156 and consists of the following:

Page No.	*Change No.	Page No.	*Change No.
Title.....	0	N-13	0
Blank.....	0	N-14 - N-18	3
A - B	2	E-1 - E-3	0
i - ii	0	E-4.....	1
iii	3	E-5.....	0
iv Blank.....	0	E-6.....	1
N-1	0	E-7.....	0
N-2	2	E-8 Blank.....	0
N-3 - N-4	3	E-9 - E-11	0
N-4.1	3	E-12 Blank.....	0
N-4.2 Blank	3	E-13 - E-14	3
N-5 - N-9	1	E-15 - E-19	0
N-10 - N-11	2	E-20.....	1
N-12 Blank	0	E-21.....	0

*** Zero in this column indicates an original page.**

Page No.	*Change No.	Page No.	*Change No.
E-22 Blank.....	0	P-22.1.....	1
E-23 - E-29	0	P-22.2 Blank	1
E-30.....	2	P-23 - P-27.....	0
E-31.....	0	P-28 - P-30.....	1
E-32 - E-35	1	P-31 - P-37.....	0
E-36 Blank.....	0	P-38 Blank	0
E-37.....	0	P-39.....	0
E-38 Blank.....	0	P-40 Blank	0
P-1 - P-2.....	0	P-41 - P-58.....	0
P-3.....	3	P-59 - P-65.....	1
P-4 - P-8.....	0	P-66 Blank	0
P-9.....	1	P-67.....	1
P-10 - P-19.....	0	P-68 Blank	0
P-20 - P-22.....	1	P-69 - P-86.....	1

*** Zero in this column indicates an original page.**

GENERAL INFORMATION AND SCOPE

SCOPE. This checklist contains the operator's checks to be accomplished during normal and emergency operations.

GENERAL INFORMATION.

NOTE

This checklist does not replace the amplified version of the procedures in the operator's manual (TM 1-1520-237-10), but is a condensed version of each procedure.

This checklist consists of three parts: normal procedures, emergency procedures, and performance data. Normal procedures consist of the procedures required for normal flight and those required for "Before Landing". The normal procedures portion will be subdivided to include the before landing checks of Chapter 8 of the operator's manual. Emergency procedures are subdivided into 10 classifications as follows: engine, propeller/rotor (insert either PROP or ROTOR), fire, fuel, electrical (ELECT), hydraulic (HYD), landing and ditching (LDG/DTCH), flight controls (FLT CONT), bailout or ejection (BAILOUT) (EJECT), and mission equipment (MSN/EQPT) (as applicable). Detailed procedures consist of detailed procedural checks.

NORMAL PROCEDURES PAGES. The contents of the normal procedures of this manual are a condensed version of the amplified checklist appearing in the normal procedures or crew duties portion of the applicable operator's manual.

EMERGENCY PROCEDURES PAGES. The requirements of this section of the condensed checklist manual (CL) are identical to those for the normal procedures, except that the information is drawn from the amplified checks in the emergency procedures portion of the operator's manual. The emergency requirements are subdivided into 10 classifications listed above. Immediate action items shall be underlined.

CHECKLIST SYMBOLS. Symbols preceding numbered steps:

- * — Indicates performance of steps is mandatory for all thru flights.
- N — Indicates performance of step is mandatory for night flights.
- ★ — Indicates a detailed procedure for this step is included in the performance checks section, located at the back of the checklist.
- I — Indicates mandatory check for instrument flights.
- O — Indicates if installed.
- ④ — Pilot not on the controls duties.
- Immediate action emergency items are underlined.

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), directly to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also send in your comments electronically to our E-mail address: 2028@redstone.army.mil or by fax 256-842-6546/DSN 788-6546. For the World Wide Web use: <https://amcom2028.redstone.army.mil>. Instructions for sending an electronic 2028 may be found at the back of the applicable Operator's manual immediately preceding the hard copy 2028.

BEFORE EXTERIOR CHECK.

1. Publications - Check.
- *2. Helicopter covers, locking devices, tiedowns, and grounding cables - Removed and secured.
- *3. Fuel - Check quantity as required.
4. Fuel sample (main and external) - Check.

EXTERIOR CHECK.

NOSE SECTION (AREA 1).

- *1. Main rotor blades - Check.
2. Fuselage - Nose area, check.

COCKPIT - LEFT SIDE (AREA 2).

1. Cockpit area - Check.
- *2. Left engine oil level - Check.
- *3. Check main landing gear drag beam for cracks.

CABIN TOP (AREA 3).

1. Cabin top - Check.
2. APU - Check; oil level, use dipstick.
- O 3. APU IPS - Check.
4. Gust lock - Check.
5. Main transmission - Check; oil level, use dipstick.

- *6. Main rotor system - Check.

INTERIOR CABIN (AREA 4).

1. Cabin - Check.
2. APU accumulator pressure gage - Check minimum 2,800 psi.
3. Transmission oil filter impending bypass indicator - Check.
4. Cargo hook - Check.
5. Survival gear and mission equipment - Check as required.

FUSELAGE - LEFT SIDE (AREA 5).

1. Fuselage - Check.
- *2. Intermediate gear box - Check; oil level.

TAIL PYLON (AREA 6).

1. Tail pylon - Check.
- *2. Tail rotor - Check.
- *3. Tail rotor gear box - Check; oil level.

FUSELAGE - RIGHT SIDE (AREA 7).

1. Fuselage - Check.

COCKPIT - RIGHT SIDE (AREA 8).

- *1. Right engine oil level - Check.
- *2. Check main landing gear drag beam for cracks.

3. Cockpit area - Check.
4. Ensure all access covers and maintenance platforms are closed and checked.
- ★*5. Crew and passenger briefing - Complete as required.

BEFORE STARTING ENGINES.

- *1. Copilot's collective - Extended and locked.
2. Shoulder harness locks - Check.
3. **PARKING BRAKE** - Release, then set.
- ★4. Circuit breakers and switches - Set.

COCKPIT EQUIPMENT CHECKS.

- *1. **FUEL PUMP** switch - **APU BOOST**.
- *2. **APU CONTR** switch - **ON**.
- *3. **APU** generator switch - **ON**.
- *4. **EXT PWR** switch - **OFF** and cable disconnected.
- O★5. **ERFS** **AUXILIARY FUEL MANAGEMENT** control panel - **TEST**.
- O*6. **ERFS** **AUXILIARY FUEL MANAGEMENT** control panel - Set fuel as required.
- *7. **EH** **IINS SYSTEMS SELECT** switches - **DG** and **VG**.
- ★*8. **EH** **IINS** - Align.

9. Caution/advisory/warning panels - Check as required.
10. **701C** **701D/CC** DEC engine fault indicator codes - Check for signal validation as required.
- N 11. Interior/exterior lighting - Set.
- *12. Avionics - On.
- *13. Doppler/GPS - Program.
- 13.1. Clocks - Set and running.
- O★14. Mission equipment - Check.
- ★*15. Cold weather control exercise - Check if temperature is below -17°C (1°F).
- *16. **AFCS FAILURE ADVISORY** lights - If on, **POWER ON RESET**.
- *17. **SAS1** off, **SAS2**, **TRIM**, **FPS**, and **BOOST** switches - Push **ON**.
- ★18. Flight controls - Check first helicopter flight of day.
- *19. Cyclic centered. Collective, from full down position, raise no more than 1 inch (to prevent droop stop pounding) and friction.
- *20. **COMPASS** switch **AHRS** AHRS Mode - **SLAVED**. Set as required.
21. **BACKUP HYD PUMP** switch - **OFF**.
- ★22. Stabilator - Check.
- *23. **GPS/DPLR NAV MODE** selector - **OFF**.

- 24. Barometric altimeters - Set.
- 24.1. Attitude reference - Check.
- O★25. Blade deice system - Test as required.

26. Avionics - Check and set as required.

O ★ *27. **AFMS** **CEFS** Auxiliary Fuel Management Panel - **TEST**, set as required.

STARTING ENGINES.

*1. **ENG FUEL SYS** selector(s) - As required.
XFD for first start of day.

*2. **ENGINE IGNITION** switch - **ON**.

*3. **GUST LOCK** caution - Verify does not appear.

*4. Fire guard - Posted if available.

*5. Rotor blades - Check clear.

★* (6.) Engine(s) - Start.

* (7.) If single-engine start was made, repeat step 6 for other engine.

* (8.) Systems - Check.

* (9.) **BACKUP HYD PUMP** switch - **AUTO**.

★ (10.) Hydraulic leak test system - Check.

★ (11.) Tail rotor servo transfer - Check.

O (12.) Deleted.

* 13. Engine warmup - Check if temperature is below -17°C (1°F).

ENGINE RUNUP.

*1. Flight controls - Hold.

- * ② **ENG POWER CONT** lever(s) - **FLY**.
- *3. Droop stops - Check out 70% to 75% **RPM R**.
- * ④ **#1 and #2 GEN** cautions - Disappear.
- I O ④1. **AUX CABIN HEATER** switch - As desired.
- *5. **EH ECS** panel switches - As desired.
- O★ ⑥ **DEICE EOT** - Check as required.
- *7. % **TRQ 1 and 2** - Matched within 5%.
- * ⑧ **EH Q/F PWR** switch - As desired.
- * ⑨ **FUEL PUMP** switch - **OFF**.
- * ⑩ **APU CONTR** switch - **OFF**.
- * ⑪ **AIR SOURCE HEAT/START** switch - As required.
- * ⑫ **ENG FUEL SYS** selectors - As required.
- * ⑬ **SAS 1 - ON**.
- *14. **GPS/DPLR NAV MODE** selector - As desired.
- *15. Collective friction - As required.
- N O*16. **HUD** - Adjust brightness, mode, barometric altitude, pitch, and roll as necessary.
- O * ⑰ **EH IINS NAVRDY** light flashing - CDU mode select switch to **NAV**.
- O * ⑱ **EH IINS SYSTEMS SELECT** switches - **IINS**.

- ①9. Engine Health Indicator Test (HIT)/Anti-Icing Check - Accomplish.
- * ②0. **FUEL BOOST PUMP CONTROL** switches - As required. Indicator lights check on, if required.
- O ★ ②1. **ERFS AFMS CEFS** External extended range fuel transfer - Check.
- * ②2. Microclimate cooling system (if use is anticipated) - Check. Turn BCA MCU control knob clockwise and check for leaks; note change in temperature. Adjust to desired cooling position.

BEFORE TAXI.

- O*1. Ejector rack lock levers unlocked.
- O*2. Volcano jettison safety pins - Remove and red arming levers to arm.
- O*3. Chaff, flare electronic module(s) safety pin(s) - Remove.
- *4. Chocks - Removed.
- *5. Doors - Secure.
- *6. **PARKING BRAKE** - Release.
- * ⑦. **TAIL WHEEL** switch - As required.
- 8. Wheel brakes - Check as required.

HOVER CHECK.

- ①. Systems - Check cautions/advisories, CDU, and PDU(s) for normal indications.

- ② Flight instruments - Check as required.
- ③ Power - Check. The power check is done by comparing the indicated torque required to hover with the predicted values from performance charts.

BEFORE TAKEOFF.

- * ① **ENG POWER CONT** levers - **FLY**.
- * ② Systems - Check.
- * ③ Avionics - As required.
- * ④ Crew, passengers, and mission equipment - Secure.

- O ⑤ **ERFS AFMS CEFS AUXILIARY FUEL MANAGEMENT** control panel - Set as required.

AFTER TAKEOFF.

- O★ ① **ERFS AFMS CEFS** Extended range fuel system transfer - As required.
- ② **EH ASE** - As required.
- O★ ③ Volcano mine launch, post mine launch - As required.

BEFORE LANDING.

- ① **TAIL WHEEL** switch - As required.
- ② **PARKING BRAKE** - As required.
- ★ ③ Crew, passengers, and mission equipment - Secure.

- O ④. **ERFS AFMS CEFS AUXILIARY FUEL MANAGEMENT** control panel switches - **OFF**.

AFTER LANDING CHECK.

- ①. **TAIL WHEEL** switch - As required.
- ②. Exterior lights - As required.
- ③. Avionics/mission equipment - As required.

PARKING AND SHUTDOWN.

- ①. **TAIL WHEEL** switch - As required.
- ②. **PARKING BRAKE** - Set.
- ③. **FUEL BOOST PUMP CONTROL** switches - **OFF**.
- 4. Landing gear - Chocked.
- O5. Volcano red arming levers - **SAFE** and jettison safety pins install.
- O 6. Ejector rack locking levers - Locked.
- O 7. Chaff, flare electronic module(s) safety pin(s) - Install.
- ⑧. **EH IINS SYSTEMS SELECT** switches - **DG/ VG**.
- O ⑨. **EH IINS** - **OFF**.
- O ⑩. **EH ECS** panel switches - **OFF**.
- O ⑪. **AUX CABIN HEATER** switch - **OFF**.

- ⑫. **SAS 1 - Off.**
- ⑬. **GPS/DPLR NAV MODE selector - OFF.**
- ⑭. **DEICE, PITOT, ANTI-ICE, HEATER and EH Q/F PWR switches - OFF.**
- ⑮. **AIR SOURCE HEAT/START switch - APU.**
- ⑯. **FUEL PUMP switch - APU BOOST.**
- ⑰. **APU CONTR switch - ON.** The **APU ON**, **BACK-UP PUMP ON**, and **APU ACCUM LOW** advisories should appear.
18. Collective raise no more than 1 inch.
19. Flight controls - Hold.
- ⑰.1. **GENERATORS NO. 1 and NO. 2 switches - OFF.**
- ⑳. **ENG POWER CONT levers - IDLE.**
- ㉑. **ENGINE IGNITION switch - OFF.**
22. Cyclic - As required to prevent anti-flap pounding.
23. Droop stops - Verify in, about 50% **RPM R**.
- ㉒. **BACKUP HYD PUMP switch - OFF.**
25. Stabilator - Slew to 0° after last flight of the day.
- ㉓. **BACK-UP PUMP ON advisory - Check disappears.**

- (27). **ENG POWER CONT** levers - **OFF** after 2 minutes at **Ng SPEED** of 90% or less.
- (28). **CEFS** On AFMP, **PRESS** switch - Select the external tanks for pressure refueling as desired for next flight.
- (29). **ENG FUEL SYS** selectors - **OFF**.
- (30). **TGT TEMP** - Monitor.
- 31. **701C 701D/CC** DEC torque indicator fault code - Check.
- 32. Avionics - Off.
- O33. **HUD ADJ/ON/OFF** switch - **OFF**.
- 34. Overhead switches - As required.
- 35. **APU** generator switch - **OFF**.
- 36. **FUEL PUMP** switch - **OFF**.
- 37. **APU CONTR** switch - **OFF**.
- 38. **BATT** switch - **OFF**.

BEFORE LEAVING HELICOPTER.

- 1. Walkaround - Complete, checking for damage, fluid leaks and levels.
- 2. Mission equipment - Secure.
- 3. Complete log book forms.
- 4. Secure helicopter - As required.

THRU-FLIGHT CHECKLIST BEFORE EXTERIOR CHECK.

- *1. Helicopter covers, locking devices, tiedowns, and grounding cables - Removed and secured.
- *2. Fuel - Check quantity as required.

EXTERIOR CHECK.

NOSE SECTION (AREA 1).

- *1. Main rotor blades - Check.

COCKPIT - LEFT SIDE (AREA 2).

- *1. Left engine oil level - Check.
- *2. Check main landing gear drag beam for cracks.

CABIN TOP (AREA 3).

- *1. Main rotor system - Check.

FUSELAGE - LEFT SIDE (AREA 5).

- O*1. Volcano armament system - Check.
- *2. Intermediate gear box - Check; oil level.

TAIL PYLON (AREA 6).

- *1. Tail rotor - Check.
- *2. Tail rotor gear box - Check; oil level.

FUSELAGE - RIGHT SIDE (AREA 7).

- O*1. Volcano armament system - Check.

COCKPIT - RIGHT SIDE (AREA 8).

- *1. Right engine oil level - Check.
- *2. Check main landing gear drag beam for cracks.
- ★*3. Crew and passenger briefing - Complete as required.

BEFORE STARTING ENGINES.

- *1. Copilot's collective - Extended and locked.
- *2. Radar altimeter - Set. **EH** Left **LO** bug 200 feet.
- *3. **ANTICOLLISION/POSITION LIGHTS** - As required.
- *4. **EH** **Q/F PWR** switch - **OFF**.
- O*5. **EH** **ECS** panel switches - **OFF**.
- *6. **GENERATORS NO. 1** and **NO. 2** switches - **ON**.
- *7. **AIR SOURCE HEAT/START** switch - **APU** (**OFF** for external air source).
- *8. Deleted.
- *9. **BATT** switch - **ON**.

COCKPIT EQUIPMENT CHECKS.

- *1. **FUEL PUMP** switch - **APU BOOST**.
- *2. **APU CONTR** switch - **ON**.
- *3. **APU** generator switch - **ON**.

- *4. **EXT PWR** switch - **OFF** and cable disconnected.
- O*5. **ERFS** **AUXILIARY FUEL MANAGEMENT** control panel - Set fuel as required.
- O*6. **EH** **IINS SYSTEMS SELECT** switches - **DG** and **VG**.
- ★*7. **EH** **IINS** - Align.
- *8. Avionics - On.
- *9. Doppler/GPS - Program.
- ★*10. Cold weather control exercise - Check if temperature is below -17°C (1°F).
- *11. **AFCS FAILURE ADVISORY** lights - If on, **POWER ON RESET**.
- *12. **SAS1** off, **SAS2**, **TRIM**, **FPS**, and **BOOST** switches - Push **ON**.
- *13. Cyclic centered. Collective, from full down position, raise no more than 1 inch (to prevent droop stop pounding) and friction.
- *14. **COMPASS** switch **AHRS** **AHRS Mode - SLAVED**. Set as required.
- *15. **GPS/DPLR NAV MODE** selector - **OFF**.
- O ★ *16. **AFMS** **CEFS** Auxiliary Fuel Management Panel - **TEST**, set as required.

STARTING ENGINES.

- *1. **ENG FUEL SYS** selector(s) - As required. **XFD** for first start of day.

- *2. **ENGINE IGNITION** switch - **ON**.
- *3. **GUST LOCK** caution - Verify does not appear.
- *4. Fire guard - Posted if available.
- *5. Rotor blades - Check clear.
- ★* (6.) Engine(s) - Start.
- * (7.) If single-engine start was made, repeat step 6 for other engine.
- * (8.) Systems - Check.
- * (9.) **BACKUP HYD PUMP** switch - **AUTO**.
- *10. Engine warmup - Check if temperature is below -17°C (1°F).

ENGINE RUNUP.

- *1. Flight controls - Hold.
- * (2.) **ENG POWER CONT** lever(s) - **FLY**.
- *3. Droop stops - Check out 70% to 75% **RPM R**.
- * (4.) **#1** and **#2 GEN** cautions - Disappear.
- *5. **EH** ECS panel switches - As desired.
- *6. % **TRQ 1** and **2** - Matched within 5%.
- * (7.) **EH** **Q/F PWR** switch - As desired.
- * (8.) **FUEL PUMP** switch - **OFF**.
- * (9.) **APU CONTR** switch - **OFF**.

- * ⑩ **AIR SOURCE HEAT/START** switch - As required.
- * ⑪ **ENG FUEL SYS** selectors - As required.
- * ⑫ **SAS 1 - ON.**
- *13. **GPS/DPLR NAV MODE** selector - As desired.
- *14. Collective friction - As required.
- N O*15. HUD - Adjust brightness, mode, barometric altitude, pitch, and roll as necessary.
- O * ⑬ **EH IINS NAVRDY** light flashing - CDU mode select switch to **NAV**.
- O * ⑭ **EH IINS SYSTEMS SELECT** switches - **IINS**.
- * ⑮ **FUEL BOOST PUMP CONTROL** switches - As required. Indicator lights check on, if required.
- * ⑯ Microclimate cooling system (if use is anticipated) - Check. Turn BCA MCU control knob clockwise and check for leaks; note change in temperature. Adjust to desired cooling position.

BEFORE TAXI.

- O*1. Ejector rack lock levers unlocked.
- O*2. Volcano jettison safety pins - Remove and red arming levers to arm.
- O*3. Chaff, flare electronic module(s) safety pin(s) - Remove.
- *4. Chocks - Removed.

*5. Doors - Secure.

*6. **PARKING BRAKE** - Release.

* (7.) **TAIL WHEEL** switch - As required.

BEFORE TAKEOFF.

* (1.) **ENG POWER CONT** levers - **FLY**.

* (2.) Systems - Check.

* (3.) Avionics - As required.

* (4.) Crew, passengers, and mission equipment - Secure.

Section I. Aircraft Systems

ENGINE.

SINGLE-ENGINE FAILURE.

1. Collective - Adjust to maintain % RPM R.

- ②. External cargo/stores - Jettison (if required).

If continued flight is not possible:

3. LAND AS SOON AS POSSIBLE.

If continued flight is possible:

4. Establish single-engine airspeed.
5. LAND AS SOON AS PRACTICABLE.

DUAL-ENGINE FAILURE.

AUTOROTATE.

DECREASING % RPM R.

1. Collective - Adjust to control % RPM R.

- ②. ENG POWER CONT lever - LOCKOUT low % **TRQ/TGT TEMP** engine. Maintain % **TRQ** approximately 10% below other engine.

3. LAND AS SOON AS PRACTICABLE.

INCREASING % RPM R.

- ①. ENG POWER CONT lever - Retard high % **TRQ/TGT TEMP** engine, maintain % **TRQ** approximately 10% below other engine.
2. LAND AS SOON AS PRACTICABLE.

If the affected engine does not respond to **ENG POWER CONT** lever movement in the range between **FLY** and **IDLE**, the HMU may be malfunctioning internally.

If this occurs:

3. Establish single engine airspeed.
- ④. Perform EMER ENG SHUTDOWN (affected engine).
- ⑤. Refer to single-engine failure emergency procedure.

% RPM INCREASING/DECREASING (OSCILLATION).

1. Slowly retard the **ENG POWER CONT** lever on the suspected engine.

If the oscillation stops:

2. Place that engine in **LOCKOUT** and manually control the power.
3. LAND AS SOON AS PRACTICABLE.

If the oscillation continues:

4. Place the **ENG POWER CONT** lever back to **FLY** and retard the **ENG POWER CONT** lever of the other engine.

When the oscillation stops:

5. Place the engine in **LOCKOUT**, manually control the power.
6. LAND AS SOON AS PRACTICABLE.

% TRQ SPLIT BETWEEN ENGINES 1 AND 2.

- ①. If **TGT TEMP** of one engine exceeds the limiter (**700** 852°C, **701C** **701D/CC** 875°C with low power engine above 50% **TRQ** or 901°C with low power engine below 50% **TRQ**), retard **ENG POWER CONT** lever on that engine to reduce **TGT TEMP**. Retard the **ENG POWER CONT** lever to maintain torque of the manually controlled engine at approximately 10% below the other engine.
- ②. If **TGT TEMP** limit on either engine is not exceeded, slowly retard **ENG POWER CONT** lever on high % **TRQ** engine and observe % **TRQ** of low power engine.
- ③. If % **TRQ** of low power engine increases, **ENG POWER CONT** lever on high power engine - Retard to maintain % **TRQ** approximately 10% below other engine (the high power engine has been identified as a high side failure).
- ④. If % **TRQ** of low power engine does not increase, or % **RPM R** decreases, **ENG POWER CONT** lever - Return high power engine to **FLY** (the low power engine has been identified as a low side failure).
- ⑤. If additional power is required, low power **ENG POWER CONT** lever, momentarily move to **LOCKOUT** and adjust to set % **TRQ** approximately 10% below the other engine.
6. LAND AS SOON AS PRACTICABLE.

ENGINE COMPRESSOR STALL.

1. Collective- Reduce.

If condition persists:

2. ENG POWER CONT lever (affected engine) - Retard (**TGT TEMP** should decrease).
3. **ENG POWER CONT** lever (affected engine) - **FLY.**

If stall condition recurs:

4. Establish single engine airspeed.
5. EMER ENG SHUTDOWN (affected engine).
6. Refer to single-engine failure emergency procedure.

ENGINE OIL FILTER BYPASS CAUTION APPEARS, ENGINE CHIP CAUTION APPEARS, ENG OIL PRESS HIGH/LOW, ENGINE OIL TEMP HIGH, ENGINE OIL TEMP CAUTION APPEARS, ENGINE OIL PRESS CAUTION APPEARS.

1. **ENG POWER CONT** lever - Retard to reduce torque on affected engine.

If oil pressure is below minimum limits or if oil temperature remains above maximum limits:

2. Establish single engine airspeed.
3. EMER ENG SHUTDOWN (affected engine).
4. Refer to single-engine failure emergency procedure.

ENGINE HIGH-SPEED SHAFT FAILURE.

1. Collective- Adjust.
2. Establish single engine airspeed.
3. EMER ENG SHUTDOWN (affected engine).
Do not attempt to restart.
4. Refer to single-engine failure emergency procedure.

LIGHTNING STRIKE.

- ①. ENG POWER CONT levers - Adjust as required to control % RPM.
2. LAND AS SOON AS POSSIBLE.

ROTORS, TRANSMISSIONS, AND DRIVE SYSTEMS.

LOSS OF TAIL ROTOR THRUST.

1. AUTOROTATE.
2. ENG POWER CONT levers - OFF (when intended point of landing is assured).

LOSS OF TAIL ROTOR THRUST AT LOW AIRSPEED/HOVER.

1. Collective- Reduce.
2. ENG POWER CONT levers - OFF (5 to 10 feet above touchdown).

TAIL ROTOR QUADRANT CAUTION AP- PEARS WITH NO LOSS OF TAIL ROTOR CONTROL.

LAND AS SOON AS PRACTICABLE.

TAIL ROTOR QUADRANT CAUTION AP- PEARS WITH LOSS OF TAIL ROTOR CONTROL.

1. Collective- Adjust.
2. LAND AS SOON AS PRACTICABLE.

PEDAL BIND/RESTRICTION OR DRIVE WITH NO ACCOMPANYING CAUTION.

1. Apply pedal force to oppose the drive.

② TRIM switch - Off.

If normal control forces are not restored:

③ BOOST switch - Off.

If control forces, normal for boost off flight are not restored:

④ BOOST switch - ON.

⑤ **TAIL SERVO** switch - **BACKUP**, if tail rotor is not restored.

6. LAND AS SOON AS PRACTICABLE.

#1 TAIL RTR SERVO CAUTION APPEARS AND BACK-UP PUMP ON ADVISORY DOES NOT APPEAR OR #2 TAIL RTR SERVO ON ADVISORY DOES NOT APPEAR.

① **TAIL SERVO** switch - **BACKUP**.

② **BACKUP HYD PUMP** switch - **ON**.

3. LAND AS SOON AS PRACTICABLE.

**MAIN XMSN OIL PRESS CAUTION
APPEARS/XMSN OIL PRESS LOW/XMSN
OIL TEMP HIGH OR XMSN OIL TEMP CAU-
TION APPEARS.**

1. LAND AS SOON AS POSSIBLE.

If time permits:

2. Slow to 80 KIAS.
- ③. EMER APU START.
- ④. **GENERATORS NO. 1 and NO. 2** switches -
OFF.

**CHIP INPUT MDL LH OR RH CAUTION AP-
PEARS.**

- ①. ENG POWER CONT lever on affected engine
- IDLE.
2. LAND AS SOON AS POSSIBLE.

**CHIP MAIN MDL SUMP, CHIP ACCESS MDL
LH OR RH, CHIP TAIL XMSN OR CHIP INT
XMSN/TAIL XMSN OIL TEMP OR INT XMSN
OIL TEMP CAUTION APPEARS.**

LAND AS SOON AS POSSIBLE.

MAIN TRANSMISSION FAILURE.

1. Collective - Adjust only enough to begin a
descent with power remaining applied to the
main transmission throughout the descent and
landing.
2. LAND AS SOON AS POSSIBLE.

FIRE.

ENGINE/FUSELAGE FIRE ON GROUND.

- ①. ENG POWER CONT levers - OFF.
- ②. ENG EMER OFF handle - Pull if applicable.
- ③. FIRE EXTGH switch - MAIN/RESERVE as required.

APU COMPARTMENT FIRE.

- ①. APU fire T-handle - Pull.
- ②. FIRE EXTGH switch - MAIN/RESERVE as required.

APU OIL TEMP HI CAUTION APPEARS.

APU CONTR switch - OFF. Do not attempt restart until oil level has been checked.

ENGINE FIRE IN FLIGHT.

1. Establish single engine airspeed.
- ②. ENG POWER CONT lever (affected engine) - OFF.
- ③. ENG EMER OFF handle - Pull.
- ④. FIRE EXTGH switch - MAIN/RESERVE as required.
5. LAND AS SOON AS POSSIBLE.

ELECTRICAL FIRE IN FLIGHT.

- ①. BATT and GENERATORS switches - OFF.
2. LAND AS SOON AS POSSIBLE.

SMOKE AND FUME ELIMINATION.

1. Airspeed - 80 KIAS or less.
2. Cabin doors and gunner's windows - Open.
3. Place helicopter out of trim.
4. LAND AS SOON AS PRACTICABLE.

FUEL SYSTEM.

#1 OR #2 FUEL FLTR BYPASS CAUTION APPEARS.

- ①. **ENG FUEL SYS** selector on affected engine - XFD.
2. LAND AS SOON AS PRACTICABLE.

#1 AND #2 FUEL FLTR BYPASS CAUTIONS APPEAR.

LAND AS SOON AS POSSIBLE.

#1 FUEL LOW AND #2 FUEL LOW CAUTIONS APPEAR.

LAND AS SOON AS PRACTICABLE.

#1 OR #2 FUEL PRESS CAUTION AP- PEARS.

If the caution appears and the situation is critical:

- ①. FUEL BOOST PUMP CONTROL switches -
NO. 1 PUMP and NO. 2 PUMP - ON.
2. LAND AS SOON AS PRACTICABLE.

If the situation is not critical:

- ①. ENG FUEL SYS selector on affected engine -
XFD.

If caution still appears:

- ②. FUEL BOOST PUMP CONTROL switches -
NO. 1 PUMP and NO. 2 PUMP - ON.

If caution still appears:

- ③. FUEL BOOST PUMP CONTROL switches -
NO. 1 PUMP and NO. 2 PUMP - OFF.
4. LAND AS SOON AS PRACTICABLE.

#1 AND #2 FUEL PRESS CAUTIONS AP- PEAR.

LAND AS SOON AS POSSIBLE.

ELECTRICAL SYSTEM.

#1 AND #2 GENERATOR FAILURE (#1 AND #2 CONV AND AC ESS BUS OFF CAUTIONS APPEAR).

- ①. **SAS 1** switch - Press off.
2. Airspeed - Adjust (80 KIAS or less).
- ③. **GENERATORS NO. 1** and **NO. 2** switches - **RESET**; then **ON**.

If cautions still appear:

- ④. **GENERATORS NO. 1** and **NO. 2** switches - **OFF**.
- ⑤. EMER APU START.
- ⑥. **SAS 1** switch - **ON**.
7. If MCUs are in use: BCA MCU control knobs (pilot, copilot, and crewmembers) - **OFF**.
8. **LAND AS SOON AS PRACTICABLE**.

#1 OR #2 GEN CAUTION APPEARS.

- ①. Affected **GENERATORS** switch - **RESET**; then **ON**.

If caution still appears:

- ②. Affected **GENERATORS** switch - **OFF**.

#1 AND #2 CONV CAUTIONS APPEAR.

- ① Unnecessary dc electrical equipment - OFF.
2. LAND AS SOON AS PRACTICABLE.

BATTERY FAULT CAUTION APPEARS.

- ① **BATT** switch - **OFF**; then **ON**. If **BATTERY FAULT** caution appears, cycle **BATT** switch no more than two times.

If caution still appears:

- ② **BATT** switch - **OFF**.

BATT LOW CHARGE CAUTION APPEARS.

If caution appears after ac power is applied:

- ① **BATT** switch - **OFF**; then **ON**. About 30 minutes may be required to recharge battery.

If caution appears in flight:

- ② **BATT** switch - **OFF**, to conserve remaining battery charge.

HYDRAULIC SYSTEM.

#1 HYD PUMP CAUTION APPEARS.

- ①. **TAIL SERVO** switch - **BACKUP**; then **NORMAL**.
2. **LAND AS SOON AS PRACTICABLE.**

#2 HYD PUMP CAUTION APPEARS.

- ①. **POWER ON RESET** switches - Simultaneously press, then release.
2. **LAND AS SOON AS PRACTICABLE.**

#1 AND #2 HYD PUMP CAUTIONS APPEAR.

LAND AS SOON AS POSSIBLE. Restrict control movement to moderate rates.

#1 OR #2 HYD PUMP CAUTION APPEARS AND BACK-UP PUMP ON ADVISORY DOES NOT APPEAR.

1. Airspeed -Adjust to a comfortable airspeed.
- ②. **BACKUP HYD PUMP** switch - **ON**.

If **BACK-UP PUMP ON** advisory still does not appear:

- ③. **FPS** and **BOOST** switches - Off (for **#2 HYD PUMP**caution).
4. LAND AS SOON AS POSSIBLE.

#1 OR #2 PRI SERVO PRESS CAUTION APPEARS.

1. SVO OFF switches - Check both are centered.
2. If the caution appears with the SVO OFF switch centered, move the SVO OFF switch to turn off the malfunctioning servo.
3. LAND AS SOON AS POSSIBLE.

#1 RSVR LOW AND #1 HYD PUMP CAUTIONS APPEAR WITH BACK-UP PUMP ON ADVISORY APPEARING.

1. LAND AS SOON AS PRACTICABLE.

If the **BACK-UP RSVR LOW** caution also appears:

- ② SVO OFF switch - 1ST STG.
3. LAND AS SOON AS POSSIBLE.

#2 RSVR LOW AND #2 HYD PUMP CAUTIONS APPEAR WITH BACK-UP PUMP ON ADVISORY APPEARING.

- ① **POWER ON RESET** switches - Simultaneously press then release.
2. LAND AS SOON AS PRACTICABLE.

If **BACK-UP RSVR LOW** caution also appears:

- ③ SVO OFF switch - 2ND STG.
4. LAND AS SOON AS POSSIBLE.

#2 RSVR LOW CAUTION APPEARS.

- ① **BOOST** and **FPS** switches- Off.
2. LAND AS SOON AS PRACTICABLE.

COLLECTIVE BOOST SERVO HARDOVER/ POWER PISTON FAILURE.

- ① **BOOST** switch - Off.
2. LAND AS SOON AS PRACTICABLE.

PITCH BOOST SERVO HARDOVER.

- ① **SAS (1 and 2)** and **FPS** switches- Off.
2. LAND AS SOON AS PRACTICABLE.

BOOST SERVO OFF CAUTION APPEARS.

- ① **BOOST** switch - Off.
2. LAND AS SOON AS PRACTICABLE.

LANDING AND DITCHING.

EMERGENCY LANDING IN WOODED AREAS - POWER OFF.

1. AUTOROTATE. Decelerate helicopter to stop all forward speed at treetop level.
2. Collective adjust to maximum before main rotor contacts tree branches.

DITCHING - POWER ON.

1. Approach to a hover.
2. Cockpit doors jettison and cabin doors open prior to entering water.
3. Pilot shoulder harness - Lock.
4. Survival gear - Deploy.
5. Personnel, except pilot, exit helicopter.
6. Fly helicopter downwind a safe distance and hover.
7. **ENG POWER CONT** levers - **OFF**.
8. Perform hovering autorotation, apply full collective to decay rotor RPM as helicopter settles.
9. Position cyclic in direction of roll.
10. Exit when main rotor has stopped.

DITCHING - POWER OFF.

1. AUTOROTATE.
2. Cockpit doors jettison and cabin doors open prior to entering water.
3. Cyclic - Position in direction of roll.
4. Exit when main rotor has stopped.

FLIGHT CONTROL/MAIN-ROTOR SYSTEM MALFUNCTIONS.

If the main rotor system malfunctions:

1. LAND AS SOON AS POSSIBLE.
- ②. EMER ENG(S) SHUTDOWN after landing.

SAS FAILURE WITH NO FAILURE ADVISORY INDICATION.

- ①. SAS 1 switch - Off.

If condition persists:

- ②. SAS 1 switch - ON.
- ③. SAS 2 switch - Off.

If malfunction still persists:

- ④. SAS 1 and FPS switches - Off.

SAS 2 FAILURE ADVISORY LIGHT ON.

POWER ON RESET switches -
Simultaneously press and then release.

SAS OFF CAUTION APPEARS.

FPS switch - Off.

FLT PATH STAB CAUTION APPEARS.

- ①. **EH** SYSTEMS SELECT - DG/VG.
- ②. **POWER ON RESET** switches -
Simultaneously press and then release.

If failure returns, control affected axis manually.

If the airspeed fault light remains illuminated on the AFCS panel:

3. Manually slew stabilator - Adjust to 0° if above 40 KIAS. The preferred method of manually slewing the stabilator up is to use the cyclic mounted stabilator slew-up switch.
4. LAND AS SOON AS PRACTICABLE.

PITCH, ROLL, OR YAW/TRIM HARDOVER.

POWER ON RESET switches -
Simultaneously press and then release.

If failure returns, control affected axis manually.

TRIM ACTUATOR JAMMED.

LAND AS SOON AS PRACTICABLE.

STABILATOR MALFUNCTION - AUTO MODE FAILURE.

1. Cyclic mounted stabilator slew-up switch- Adjust if necessary to arrest or prevent nose down pitch rate.
2. **AUTO CONTROL** switch - Press **ON** once after establishing a comfortable airspeed.

If automatic control is not regained:

3. Manually slew stabilator - Adjust to 0° for flight above 40 KIAS or full down when airspeed is below 40 KIAS. The preferred method of manually slewing the stabilator up is to use the cyclic mounted stabilator slew-up switch.
4. **LAND AS SOON AS PRACTICABLE.**

If manual control is not possible:

5. **STAB POS** indicator - Check and fly at or below **KIAS LIMITS** shown on placard.
6. **LAND AS SOON AS PRACTICABLE.**

UNCOMMANDED NOSE DOWN/UP PITCH ATTITUDE CHANGE.

If an uncommanded nose down pitch attitude occurs:

1. Cyclic - Adjust as required.
2. Collective- Maintain or increase.
3. Cyclic mounted stabilator slew-up switch - Adjust as required to arrest nose down pitch rate.
- ④ **MAN SLEW** switch - Adjust to 0° at airspeeds above 40 KIAS and full down at airspeeds below 40 KIAS.
5. LAND AS SOON AS PRACTICABLE.

If an uncommanded nose up pitch attitude occurs:

1. Cyclic - Adjust as required.
2. Collective- Reduce as required.
- ③ **MAN SLEW** switch - Adjust to 0° at airspeeds above 40 KIAS and full down at airspeeds below 40 KIAS.
4. LAND AS SOON AS PRACTICABLE.

Section II. Mission Equipment

EMERGENCY JETTISONING.

CARGO REL or HOOK EMER REL switch - Press.

EMERGENCY RELEASE OF RESCUE HOIST LOAD.

If the rescue hoist becomes jammed, inoperative, or the cable is entangled and emergency release is required:

To cut cable from cockpit:

CABLE SHEAR switch - FIRE.

To cut cable from hoist operator's position:

CABLE CUT switch - FIRE.

BLADE DEICE SYSTEM MALFUNCTIONS.

MR DE-ICE FAULT, MR DE-ICE FAIL, OR TR DE-ICE FAIL CAUTION APPEARS.

1. Icing conditions - Exit.

②. **BLADE DEICE POWER** switch - **OFF**, when out of icing conditions.

If vibrations increase:

3. LAND AS SOON AS POSSIBLE.

PWR MAIN RTR AND/OR TAIL RTR MONITOR LIGHT ON.

If a **PWR** monitor light is illuminated with **BLADE DEICE POWER** switch **ON** to stop power from being applied to blades:

1. Icing conditions - EXIT.

② **BLADE DEICE POWER** switch - **OFF**.

If a **PWR** monitor light is still illuminated with **BLADE DEICE POWER** switch **OFF**:

③ **GENERATORS NO. 1** or **NO. 2** switch - **OFF**.

④ **APU** generator switch - **OFF** (if in use).

5. LAND AS SOON AS PRACTICABLE.

ICE RATE METER FAIL OR INACCURATE.

① **BLADE DEICE MODE** switch - **MANUAL** as required.

If vibration levels increase or **%TRQ** required increases:

② **BLADE DE-ICE MODE** switch - Select higher setting.

If ice buildup continues:

3. LAND AS SOON AS PRACTICABLE.

**LOSS OF NO. 1 OR NO. 2 GENERATOR
DURING BLADE DEICE OPERATION.**

- ①. EMER APU START.
- ②. **BLADE DE-ICE POWER** switch - **OFF** then **ON.**

EXTERNAL EXTENDED RANGE FUEL SYSTEM FAILURE TO TRANSFER SYMMETRICALLY. **ERFS**

If asymmetric fuel transfer is suspected:

- ①. Stop transfer on tank set.
- ②. Select other tank set and initiate transfer.
3. LAND AS SOON AS PRACTICABLE.

Should controlled flight with one heavy external tank become necessary, proceed as follows:

1. Make all turns shallow (up to standard rate), and in the direction away from heavy side (particularly when a right tank remains full).
2. Avoid abrupt control motions, especially lateral cyclic.
3. If possible, shift personnel to the light side of the helicopter.
4. Select a suitable roll-on landing area, and make a roll-on landing with touchdown speed in excess of 30 KIAS. To increase control margin, execute the approach into the wind or with a front quartering wind from the heavy side and align the longitudinal axis of the helicopter with the ground track upon commencing the approach. If a suitable roll-on landing area is not available, make an approach to a hover into the wind, or with a front quartering wind from the heavy side.

AUXILIARY FUEL MANAGEMENT SYSTEM FAILURE TO TRANSFER SYMMETRICALLY. **AFMS** **CEFS**

When an asymmetric fuel condition **AFMS** (greater than 400-pound difference between external tank pairs) ◀ **CEFS** (270 pounds difference for outboard tanks or 375 pounds difference for inboard tanks) ◀ is identified:

1. **XFER MODE** switch - **MAN**.
2. **MAN XFER** switch - Select heavy tank **LEFT** or **RIGHT** until imbalance condition is corrected. If the imbalance condition cannot be corrected:
3. **XFER MODE** switch - **OFF**.
4. **LAND AS SOON AS PRACTICABLE**.

Should controlled flight with one heavy external tank become necessary, proceed as follows:

1. Make all turns shallow (up to standard rate), and in the direction away from heavy side (particularly when a right tank remains full).
2. Avoid abrupt control motions, especially lateral cyclic.
3. If possible, shift personnel to the light side of the helicopter.
4. Select a suitable roll-on landing area, and make a roll-on landing with touchdown speed in excess of 30 KIAS. To increase control margin, execute the approach into the wind or with a front quartering wind from the heavy side and

align the longitudinal axis of the helicopter with the ground track upon commencing the approach. If a suitable roll-on landing area is not available, make an approach to a hover into the wind, or with a front quartering wind from the heavy side.

EXTERNAL STORES JETTISON.

If jettisoning of tanks is required:

- ①. **STORES JETTISON** switch - Select **INBD BOTH, OUTBD BOTH** or **ALL** as applicable.
- ②. **JETT** switch - Actuate.

If primary jettison system does not operate:

- ③. **EMER JETT ALL** switch - Actuate.

FUEL FUMES IN COCKPIT/CABIN WITH EXTERNAL EXTENDED RANGE FUEL SYSTEM PRESSURIZED.

ERFS **AFMS** **CEFS**

If heater is on:

- ①. **HEATER** switch - **OFF**.

If heater is off or fumes persist:

- ②. **ERFS** **PRESS OUTBD** and **INBD** switches - **OFF**. ◀ **AFMS** **CEFS** **PRESS** switch - **OFF**. ◀
- ③. **AFMS** **CEFS** **XFER** ◀ **MODE** switch - **OFF**.
- ④. **FUEL BOOST PUMP CONTROL** switches -
As required.

VOLCANO LAUNCHER RACKS JETTISON.

If jettisoning of launcher rack is required:

1. **JETTISON** switch - **JETTISON**.

If jettison procedure above fails, do the following immediately:

2. **EMER JETTISON** switch - **JETTISON**.

INFLIGHT CABS ACTIVATION.

If inadvertent deployment occurs:

1. If at terrain flight altitude, initiate climb if appropriate.
2. **CABS** circuit breaker - Pull.
3. Fold the deflated forward air bag over the top of the glare shield to minimize flight control and flight instrument interference.
4. Push the lateral air bag down and rear into the space between the armor panel and door.
5. Transfer controls and repeat steps 3. and 4. for other side of helicopter.

DETAILED PROCEDURES

CREW BRIEFING.

A crew briefing shall be conducted to ensure a thorough understanding of individual and team responsibilities. The briefing should include, but not be limited to, pilots, crew chief, mission equipment operator, ground crew responsibilities, and the coordination necessary to complete the mission in the most efficient manner. A review of visual signals is desirable when ground guides do not have direct voice communications link with the crew.

PASSENGER BRIEFING.

The following guide may be used in accomplishing required passenger briefings. Items that do not pertain to a specific mission may be omitted.

1. Crew introduction.
2. Equipment.
 - a. Personal, to include ID tags.
 - b. Professional.
 - c. Survival.
3. Flight data.
 - a. Route.
 - b. Altitude.
 - c. Time en route.
 - d. Weather.
4. Normal procedures.

- a. Entry and exit the helicopter.
 - b. Seating.
 - c. Seatbelts.
 - d. Movement in helicopter.
 - e. Internal communications.
 - f. Security of equipment.
 - g. Smoking.
 - h. Oxygen.
 - i. Refueling.
 - j. Weapons.
 - k. Protective masks.
 - l. Parachutes.
 - m. Hearing protection.
 - n. Aviation life support equipment (ALSE).
5. Emergency procedures.
- a. Emergency exits.
 - b. Emergency equipment.
 - c. Emergency landing/ditching procedures.

CIRCUIT BREAKERS AND SWITCHES - SET.

- a. Circuit breakers - In.

- b. Avionics - Off, frequencies set.
- c. **BLADE DEICE POWER** switch - **OFF**.
- *d. Radar altimeter - Set. **EH** Left **LO** bug 200 feet.
- e. Deleted.
- f. **BACKUP HYD PUMP** switch - **AUTO**.
- *g. **ANTICOLLISION/POSITION LIGHTS**
- As required.
- *h. **EH Q/F PWR** switch - **OFF**.
- O*i. **EH ECS** panel switches - **OFF**.
- j. **CARGO HOOK EMERG REL** switch - **OPEN**, **ARMING** switch - **SAFE**.
- k. **APU CONTR** switch - **OFF**; **APU** T-handle - In.
- *l. **GENERATORS NO. 1** and **NO. 2** switches - **ON**.
- m. Ground power unit - Connected if required.
- *n. **AIR SOURCE HEAT/START** switch - **APU (OFF** for external air source).
- *o. Deleted.
- p. **EMER OFF** T-handles - Full forward.
- *q. **BATT** switch - **ON**.

AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST. **ERFS**

1. **TEST** button - Press. Digits should display 8s, all panel lights and **DEGRADED** and **VENT SENSOR (FAIL and OVFL)** lights should illuminate.
2. **TEST** button - Release. Digits should display 8s in sequence from left to right three times; 5 seconds later, display **GOOD** or **EO** failure code; 3 seconds later, display type fuel density, then fuel **TOTAL**.
3. Auxiliary fuel quantity switch - **CAL**.
4. **INCR/DECR** switch - Set calibration.
5. Auxiliary fuel quantity switch - **INBD**.
6. **INCR/DECR** switch - Set inboard fuel quantity.
7. Auxiliary fuel quantity switch - **OUTBD**.
8. **INCR/DECR** switch - Set outboard fuel quantity.
9. Auxiliary fuel quantity switch move to **TOTAL** - Check.
10. **PRESS OUTBD** and **INBD** switches - As desired.

EH STARTING PROCEDURE (IINS NORMAL ALIGNMENT).

1. Ensure that the circuit breakers are in.
2. **IINS** mode select switch - **NORM**.

3. Page selector switch - **POS**.
4. Verify line 7 on right side display indicates desired COORDINATE SYSTEM (UTM or L/L). If not press the line select key once to switch to the desired coordinate system.
 - a. Enter GRID ZONE/SPHEROID or LATITUDE in scratchpad.
 - b. Press line select key 5 left.
 - c. Enter AREA, EASTINGS, NORTHINGS, or LONGITUDE in scratchpad.
 - d. Press line select key 7 left.
5. Verify line 3 on left side display indicates correct magnetic variation, MV.
 - a. If incorrect, enter MV in scratchpad.
 - b. Verify scratchpad entry is correct.
 - c. Press line select key 3 left.
 - d. Verify line 3 left displays: - MV = XNN.N.
(The "=" sign indicates that a manual MV was entered and automatic MV updating will not occur.)
6. Page select switch - **INS**.
 - a. Enter barometric pressure of present position in scratchpad.
 - b. Press line select key 5 left.

- c. Enter altitude of present position in scratchpad (e.g., 156 ft is entered as 0.156 and displayed as 0.2).
 - d. Press line select key 3 left.
7. Page select switch - **DEST**.
- a. Press **DEST** toggle switch to select **DEST** desired page.
 - b. Press line select key 7 right to display desired coordinate system (**UTM** or **L/L**).
 - c. Enter grid zone and spheroid or latitude in scratchpad.
 - d. Press line select key 5 left.
 - e. Enter Area/Eastings/Northings or longitude in scratchpad.
 - f. Press line select key 7 left.
 - g. Press **DEST** toggle switch to increment to the next page.
8. Page select switch - **TCN**.
- a. Turn ON TACAN by pressing line select key 1 left.
 - b. Press line 3 left until REC is displayed on the CRT.
 - c. Press page slew toggle switch to display TACAN station zero page.
 - d. Enter magnetic variation in scratchpad.

- e. Press line select key 3 left.
 - f. Enter latitude in scratchpad.
 - g. Press line select key 5 left.
 - h. Enter longitude in scratchpad.
 - i. Press line select key 7 left.
 - j. Press line select key 1 right to display ACT.
 - k. Enter channel number in scratchpad.
 - l. Press line select key 3 right.
 - m. Enter elevation of TACAN station on scratchpad.
 - n. Press line select key 5 right.
 - o. Press page slew toggle switch to display next TACAN page.
 - p. Enter data as described in steps d. through o.
9. Select an appropriate destination number and toggle the **STR** toggle switch to indicate the number, i.e., S1. (It is not necessary that the DX and SX numbers agree, only that SX is the desired destination.)
10. Mode select switch - **NAV**. (Pull switch up; then rotate.)
11. Page select switch - **TCN**.
- a. Press line select key 3 left to display T/R.

- b. Press line select key 5 left to display UPDT.
12. On **VSI/HSI MODE SEL** panel, press **IINS** switch. Note that bearing to destination (No. 1 needle), range to destination, course deviation and **TO/FROM** flag are displayed on the HSI.
13. On **SYSTEMS SELECT** panel, set switches and observe indications as follows:
 - a. Press **HDG** switch, **IINS** illuminates and inertial derived heading is displayed on the HSI.
 - b. Press **ATT** switch, **IINS** illuminates and inertial derived pitch and roll is displayed on the VSI.

MISSION EQUIPMENT

RADAR SIGNAL DETECTING SET AN/APR-39A(V)1.

1. Self-test mode.
 - a. After power up, the AN/APR-39A(V)1 synthetic voice will announce "APR-39 POWER UP" and the (+) symbol will stabilize in the center of the CRT. Self-test should be initiated after approximately one minute. Self-test can be performed in **MODE 1** or **MODE 2**. In **MODE 1** the synthetic voice will announce "SELF-TEST SET VOLUME, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ". In **MODE 2** the synthetic voice will announce "SELF-TEST SET VOLUME, 5, 4, 3, 2, 1".
 - b. The CRT will display specific software version numbers i.e., operational flight program

(OFP) at the 12-o'clock position and the emitter identification data (EID) at the 6-o'clock position.

- c. After the software version numbers have been displayed the test sequence checks the receivers. A good visual self-test will show two triangles, one at the 6-o'clock position and one at 12-o'clock position on the CRT. This is a normal indication and does not effect system performance.
 - d. A good self-test (no faults detected) ends with the message "APR-39 OPERATION-AL". A bad self-test (faults detected) ends with the "APR-39 FAILURE".
2. **MODE 1** operation. Selecting **MODE 1** the operator will hear all the normal synthetic voice audio when an emitter has been processed e. g., the AN/APR-39A(V)1 will announce; "SA, SA-6 12-O'CLOCK TRACKING". Selection of this mode does not have any effect on emitters received, processed, or displayed; it only affects synthetic voice audio.
3. **MODE 2** operation. Selecting **MODE 2** the operator will hear an abbreviated synthetic voice audio e.g., the AN/APR-39A(V)1 will announce; "MISSILE 12-O'CLOCK TRACKING".

CARGO HANDLING SYSTEMS

OPERATIONAL CHECK - NORMAL RELEASE MODE.

1. **CARGO HOOK CONTR** switch - As required. **CKPT** for pilot and copilot check, or **ALL** for crewmember check.
2. **CARGO HOOK ARMING** switch - **ARMED**.
3. **HOOK ARMED** advisory - Check that it appears.
4. Place about 20 pounds downward pressure on load beam.
5. **CARGO REL** switch (pilot and copilot); **NORMAL RLSE** (crewmember) - Press and release.
6. Load beam - Check open. **CARGO HOOK OPEN** advisory - Appears.
7. **CARGO HOOK OPEN** advisory - Check that it disappears when hook closes.
8. Repeat steps 4. through 7. for copilot and crewmember positions.

OPERATIONAL CHECK - MANUAL RELEASE MODE.

1. Manual release lever spring - Installed. Check that spring is straight and provides positive pressure on the lever.
2. Place about 20 pounds downward pressure on load beam.

3. Manual release lever - Pull up/turn fully clockwise and release.
4. Load beam - Check open.
5. **CARGO HOOK OPEN** advisory - Appears.
6. When downward pressure is released, load beam will close and latch.
7. **CARGO HOOK OPEN** advisory - Disappears when hook closes.

CARGO HOOK EMERGENCY RELEASE CIRCUIT CHECK.

1. **EMERG REL TEST** light - Press. Light should be on.
2. Pilot's release - Check.
 - a. Short test.
 - (1) **CARGO HOOK EMERG REL** switch - **SHORT**.
 - (2) Pilot's **HOOK EMER REL** switch - Press and hold.
 - (3) **CARGO HOOK TEST** light - On.
 - (4) **HOOK EMER REL** switch - Release. **TEST** light off.
 - (5) Repeat steps (2) through (4) for copilot's **HOOK EMER REL** switch and crewmember's cargo hook control pendant **EMER RLSE** switch.

b. Open test.

- (1) **CARGO HOOK EMERG REL** switch - **OPEN**.
 - (2) Pilot's **HOOK EMER REL** switch - Press and hold.
 - (3) **CARGO HOOK TEST** light - On.
 - (4) **HOOK EMER REL** switch - Release. **TEST** light off.
 - (5) Repeat steps (2) through (4) for copilot's **HOOK EMER REL** switch and crewmember's cargo hook control pendant **EMER RLSE** switch.
3. **CARGO HOOK EMERG REL** switch - **NORM**. If the cargo hook is not to be used immediately after completing the circuit test check, the **EMERG REL** switch shall remain at **OPEN** until ready for load pickup.

RESCUE HOIST SYSTEM KIT.

RESCUE HOIST SQUIB CIRCUIT TEST.

1. **SQUIB** switch - Hold at **TEST**.
2. **SQUIB IND** light - Check on.
3. **SQUIB** switch - Release to **NORM**. **SQUIB IND** light off.

BOOM POSITION AND HOIST CABLE CONTROL OPERATIONAL CHECK.

1. **MASTER** switch - **ON**.

2. Hoist operator - Check power on indicator (blue light), check to ensure that the 10-foot **CAUTION** light on the pendant grip is on, and cooling fan operating.
3. Check ICS switch on pendant.
4. Hoist operator - **BOOM** switch - **OUT** and then **IN**.
5. **RESCUE HOIST CONTROL** panel - Rotate boom **OUT**; then **IN**, then **OUT** to test boom operation.
6. Speed mode switch - **HIGH SPEED**.
7. **RESCUE HOIST CONTROL** panel - **CABLE** switch - **DOWN**; lower cable hook and ensure the 10-foot **CAUTION** light on the pendant grip extinguishes after the first ten feet of cable is reeled out.
8. **RESCUE HOIST CONTROL** panel **CABLE** switch - **UP**; reel in cable and observe that cable speed slows when the 10-foot **CAUTION** light on the pendant grip goes on (8 to 10 feet of cable out).
9. Boom up limit actuator arm - Push up on arm during reeling in to check that hoist stops running when up limit switches are activated. Observe that cable slows when hook is 12 to 18 inches from full up position.
10. Repeat steps 7. through 9., using control pendant assembly. Check that cable speed can be regulated by control pendant from 0 to 250 fpm when cable is reeled out beyond 10 feet.

11. Speed mode switch - **LOW SPEED**. Repeat steps 7. through 9., using control pendant assembly. Check that cable speed can be regulated by control pendant from 0 to 125 fpm when cable is reeled out beyond 10 feet.
12. **BOOM** switch - Rotate boom into the stowed position.

COLD WEATHER CONTROL EXERCISE.

After starting the APU, the controls must be exercised when operating in a temperature range of -17°C (1°F) and below. The control exercise is required:

- a. At temperatures between -17°C (1°F) and -31°C (-24°F), cycle collective slowly for 1 minute.
 - (1) Move collective stick grip up about 3 inches from lower stop, and down again 30 times during 1 minute of control cycling in step a.
 - (2) Move each tail rotor pedal alternately through $\frac{3}{4}$ inch of travel from neutral position 30 times during 1 minute of control cycling in step a.
- b. At temperatures between -31°C (-24°F) and -43°C (-45°F), cycle collective slowly for 2 minutes.
 - (1) Move collective stick grip up about $1\frac{1}{2}$ inches from lower stop and down again during first minute, and 3 inches of travel during second minute of control cycling in step b.
 - (2) Move each tail rotor pedal alternately through $\frac{3}{8}$ inch of travel from neutral position during first minute and $\frac{3}{4}$ inch of travel

during second minute of control cycling in step b.

- c. At temperatures between -43°C (-45°F) and -54°C (-65°F), cycle collective slowly for 5 minutes.

Move collective and pedals through travel for times shown below:

<u>Collective Travel</u> <u>(Approximately)</u>	<u>Pedals Travel</u> <u>(Approximately)</u>	<u>Time Duration</u>
3/4 inch	1/8 inch	First minute
1 1/2 inches	1/4 inch	Second minute
1 3/4 inches	1/2 inch	Third minute
2 1/2 inches	5/8 inch	Fourth minute
3 inches	3/4 inch	Fifth minute

FLIGHT CONTROLS - CHECK FIRST HELICOPTER FLIGHT OF DAY.

- a. Collective - Midposition, pedals centered, friction off.
- b. **BOOST** switch - Press off. There will be a slight increase in collective and pedal forces. **BOOST SERVO OFF** caution and **MASTER CAUTION** should appear.
- c. Right **SVO OFF** switch - **1ST STG.** No allowable cyclic stick jump. **#1 PRI SERVO PRESS** caution and **MASTER CAUTION** should appear.
- d. Move cyclic and pedals slowly through full range. There should be no binds or restrictions. Move collective full up to full down in about 1 to 2 seconds. Check **#2 PRI**

SERVO PRESS caution does not appear during movement of collective.

- e. Right **SVO OFF** switch - **2ND STG.** No allowable cyclic stick jump. **#2 PRI SERVO PRESS** caution and **MASTER CAUTION** should appear.
- f. Repeat step d. above. Check **#1 PRI SERVO PRESS** caution does not appear during movement of collective.
- g. **SVO OFF** switch - Center.
- h. Collective - Move through full range in no less than 5 seconds. There should be no binding.
- i. Pedals - Move both pedals through the full range in no less than 5 seconds. There should be no binding.
- j. **TAIL SERVO** switch - **BACKUP.** **#1 TAIL RTR SERVO** caution, both **MASTER CAUTION** cautions, and **#2 TAIL RTR SERVO ON** advisory appear. Move pedals through full range in no less than 5 seconds. There should be no binding.
- k. **TAIL SERVO** switch - **NORMAL.** Cautions and advisories should disappear.
- l. **BOOST** switch - **ON.** **BOOST SERVO OFF** caution should disappear.

STABILATOR - CHECK.

- a. **STAB POS** indicator should be between 34° and 42° **DN.**

- b. **TEST** button - Press and hold. Check **STAB POS** indicator moves up 5° to 12°. **STABILATOR** caution and **MASTER CAUTION** appear; stabilator audio heard.
- c. **AUTO CONTROL RESET** switch - Press **ON**. Note that the **STABILATOR** caution disappears and audio is off, and **STAB POS** indicator moves to 34° to 42° down.
- d. Either cyclic mounted stabilator slew-up switch - Press and hold until **STAB POS** indicator moves approximately 15° trailing edge up, release, stabilator should stop. **STABILATOR** caution and **MASTER CAUTION** appear and beeping audible warning in pilot's and copilot's headsets. **MASTER CAUTION** - Press to reset audio tone.
- e. Other cyclic mounted stabilator slew-up switch - Press and hold until **STAB POS** indicator moves approximately 15° trailing edge up, release, stabilator should stop.
- f. **MAN SLEW** switch - **UP** and hold until stabilator stops. **STAB POS** indicator should be 6° to 10° up.
- g. **MAN SLEW** switch - **DN** and hold until **STAB POS** indicator reads 0°.
- h. **AUTO CONTROL RESET** switch- Press **ON**. **STAB POS** indicator should move 34° to 42° **DN**. **STABILATOR** caution disappears.

BLADE DEICE SYSTEM - TEST AS REQUIRED.

- a. Ice rate meter **PRESS TO TEST** button - Press and release.
- b. Ice rate meter indicator - Moves to half scale (1.0) holds about 50 seconds; then falls to 0 or below. **ICE DETECTED** caution and **MASTER CAUTION** appear after 15 to 20 seconds into the test, and **FAIL** flag should not be visible in flag window. Ice rate meter should move to zero within 75 seconds after pressing **PRESS TO TEST** button.
- c. **BLADE DE-ICE TEST** panel select switch - **NORM**.
- d. **PWR MAIN RTR** and **TAIL RTR** monitor lights - Press to test.
- e. **BLADE DEICE POWER** switch - **TEST**.
- f. **PWR MAIN RTR** and **TAIL RTR** monitor lights - Check. **MAIN RTR** monitor light may go on for 2 to 4 seconds. If either light remains on for 10 seconds or more:
 - (1) **BLADE DEICE POWER** switch - **OFF**. If either light is still on:
 - (2) **APU generator switch** and/or **EXT PWR** switch - **OFF**.
- g. **TEST IN PROGRESS** light - Check. The light should be on for 105 to 135 seconds. No other blade deice system lights should be on. **PWR MAIN RTR** and **TAIL RTR** monitor lights may go on momentarily near

end of test. The **TEST IN PROGRESS** light should then go off.

- h. Crewman touch each droop stop cam - Cams should be warm to touch.
- i. **BLADE DEICE POWER** switch - **OFF**.
- j. **BLADE DE-ICE TEST** panel select switch - **SYNC 1**.
- k. **BLADE DEICE POWER** switch - **TEST**. **MR DE-ICE FAIL** caution and **MASTER CAUTION** appear.
- l. **BLADE DEICE POWER** switch - **OFF**. **MR DE-ICE FAIL** caution and **MASTER CAUTION** disappear.
- m. **BLADE DE-ICE TEST** panel select switch - **SYNC 2**.
- n. **BLADE DEICE POWER** switch - **TEST**. **MR DE-ICE FAIL** caution and **MASTER CAUTION** appear.
- o. **BLADE DEICE POWER** switch - **OFF**. **MR DE-ICE FAIL** caution and **MASTER CAUTION** disappear.
- p. **BLADE DE-ICE TEST** panel select switch - **OAT**.
- q. **BLADE DEICE POWER** switch - **TEST**. **MR DE-ICE FAIL** caution, **TR DE-ICE FAIL** caution, and **MASTER CAUTION** appear.
- r. **BLADE DEICE POWER** switch - **OFF**. **MR DE-ICE FAIL** caution, **TR DE-ICE**

FAIL caution, and **MASTER CAUTION** disappear.

- s. **BLADE DE-ICE TEST** panel select switch
- **NORM**.

AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST. AFMS

1. **AUX FUEL QTY** digital readouts - Note current reading.
2. **TEST/RESET** button - Press. All annunciators will light and **E07** will be displayed, if test is initiated within approximately one minute of applying ac power. If **E07** appears, press **TEST/RESET** again. **E07** should not appear until test is finished.
3. **BIT** indications - Check. Ensure the **AUX FUEL** caution does not appear. Press and release **TEST/RESET** button on AFMP to reset the **AUX FUEL** caution.
4. **PRESS** switch - As desired for tanks installed.

AUXILIARY FUEL MANAGEMENT CONTROL PANEL TEST. CEFS

1. **AUX FUEL QTY** digital readouts - Note current reading.
2. **TEST/RESET** button - Press. All annunciators will light and **E07** will be displayed, if test is initiated within approximately one minute of applying ac power. If **E07** appears, press **TEST/RESET** again. **E07** should not appear until test is finished.

3. **BIT** indications - Check. Ensure the **AUX FUEL** caution does not appear. Press and release **TEST/RESET** button on AFMP to reset the **AUX FUEL** caution.

ENGINE(S) - START.

- a. If any of the emergency shutdown criteria/indicators occur, abort the start.
 - (1) No **TGT TEMP** increase (light off) within 45 seconds.
 - (2) No **ENG OIL PRESS** within 45 seconds.
 - (3) No **% RPM 1** or **2** or **% RPM R** within 45 seconds.
 - (4) **ENGINE STARTER** caution disappears before reaching 52% **Ng SPEED**.
 - (5) **TGT TEMP** reaches **700** 850°C or **701C** **701D/CC** 851°C before idle is attained (63% **Ng SPEED**).
- b. Starter button(s) - Press until **Ng SPEED** increases; release.
- c. **TGT TEMP** - Check below **700** 150°C or **701C** **701D/CC** 80°C before advancing **ENG POWER CONT** levers. **701C** **701D/CC** If engine is being started one hour or less since previous shutdown, motor the starter for 60 seconds prior to advancing **ENG POWER CONT** levers.
- d. **ENG POWER CONT** lever(s) - **IDLE**. Start clock.

- e. System indications - Check.
- f. **ENGINE STARTER** caution(s) - Check disappear at 52% to 65% **Ng SPEED**. If **ENGINE STARTER** caution remains after 65% **Ng**:
 - (1) **ENG POWER CONT** lever - Pull out.

If caution remains:

- (2) **APU CONTR** switch - **OFF** or engine air source remove as required.

HYDRAULIC LEAK TEST SYSTEM - CHECK.

- a. **HYD LEAK TEST** switch - **TEST**. #1 **TAIL RTR SERVO**, **BOOST SERVO OFF**, **SAS OFF**, #1 and #2 **RSVR LOW**, and **BACK-UP RSVR LOW** cautions, #2 **TAIL RTR SERVO ON** and **BACK-UP PUMP ON** advisories, and **MASTER CAUTION** appear. During this check it is normal for the collective and pedals to move slightly.
- b. **HYD LEAK TEST** switch - **RESET**. The cautions and advisories in a. should disappear.

TAIL ROTOR SERVO TRANSFER - CHECK.

- a. **BACKUP HYD PUMP** switch - **AUTO** with backup pump not running.
- b. **TAIL SERVO** switch - **BACKUP**. #1 **TAIL RTR SERVO** caution, #2 **TAIL RTR SERVO ON** and **BACK-UP PUMP**

ON advisories, and **MASTER CAUTION** appear within 3 to 5 seconds.

- c. **TAIL SERVO** switch - **NORMAL**. #1 **TAIL RTR SERVO** caution, #2 **TAIL RTR SERVO ON** advisory, and **MASTER CAUTION** disappear. **BACK-UP PUMP ON** advisory disappears after approximately 90 seconds.

DEICE EOT - CHECK AS REQUIRED.

- a. **BLADE DE-ICE TEST** select switch - **EOT**.
- b. **BLADE DEICE MODE** select switch - **MANUAL M**.
- c. **BLADE DEICE POWER** switch - **ON**.
- d. **TR DE-ICE FAIL** caution and **MASTER CAUTION** appear after 15 to 30 seconds, **MR DE-ICE FAIL** caution appears after

50 to 70 seconds, and **MR DE-ICE FAULT** caution may appear.

- e. **BLADE DEICE POWER** switch - **OFF**. **TR DE-ICE FAIL** caution, **MR DE-ICE FAIL** caution, and **MASTER CAUTION** disappear.
- f. **BLADE DE-ICE TEST** select switch - **NORM**.
- g. **GENERATORS NO. 1** or **NO. 2** switch - **OFF**. Applicable **GEN** caution and **MASTER CAUTION** appear.
- h. **BLADE DEICE POWER** switch - **ON**. Wait 30 seconds, no deice cautions appear.
- i. **GENERATORS** switch(es) - **ON**. Applicable **GEN** caution disappears.
- j. **BLADE DEICE POWER** switch - **OFF**.
- k. **BLADE DEICE MODE** select switch - **AUTO**.

EXTERNAL EXTENDED RANGE FUEL TRANSFER CHECK. ERFS

1. **AIR SOURCE HEAT/START** switch - **ENG**.
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS OUTBD** and **INBD** switches - **ON** for tanks installed.
4. Fuel quantity switch - **TOTAL**.
5. **TANKS** switch - **OUTBD**.

6. **MODE** switch - **MANUAL**.
7. **MANUAL XFR RIGHT** switch - **ON**.
8. **UH-60A** **TOTAL FUEL** or **UH-60L** **MAIN FUEL** readout - Check for increase of about 20 pounds.
9. **TANKS** switch - Repeat steps 7. and 8. for **IN-BD**, if installed.
10. **MANUAL XFR RIGHT** switch - **OFF**.
11. **MANUAL XFR LEFT** switch - **ON**.
12. Repeat steps 8. and 9. for **MANUAL XFR LEFT**.
13. **MANUAL XFR LEFT** switch - **OFF**.
14. External ERFS - Set as desired.

EXTERNAL EXTENDED RANGE FUEL TRANSFER IN AUTO MODE. **ERFS**

1. **AIR SOURCE HEAT/START** switch - **ENG**.
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS OUTBD** and **INBD** switches - **ON** for tanks installed.
4. **MODE** switch - **AUTO**.
5. **TANKS** switch - **OUTBD**. Switch to **INBD** after outboard tanks are empty.

EXTERNAL EXTENDED RANGE FUEL TRANSFER IN MANUAL MODE. **ERFS**

If **AUTO** mode is inoperative, transfer in **MANUAL** mode as follows:

1. **AIR SOURCE HEAT/START** switch - **ENG.**
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS OUTBD** and **INBD** switches - **ON** for tanks installed.
4. **MODE** switch - **MANUAL**.
5. **TANKS** switch - **OUTBD**. Switch to **INBD** after outboard tanks are empty.
6. **MANUAL XFR** switches **RIGHT** and **LEFT** - **ON**.

EXTERNAL EXTENDED RANGE FUEL FLOW VERIFICATION IN MANUAL MODE. **ERFS**

1. **AIR SOURCE HEAT/START** switch - **ENG.**
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS OUTBD** and **INBD** switches - **ON** for tanks installed.
4. **MODE** switch - **MANUAL**.
5. **TANKS** switch - **OUTBD**. Switch to **INBD** after outboard tanks are empty.

6. **MANUAL XFR RIGHT** switch - **ON**. Note the rate of decrease of the **AUX FUEL QTY POUNDS** indicator. The normal transfer fuel flow rate per tank should be between 20 to 38 pounds per minute.
7. **MANUAL XFR RIGHT** switch - **OFF**.
8. Repeat steps 6. and 7. for left tank.

EXTERNAL AUXILIARY FUEL MANAGEMENT SYSTEM FUEL TRANSFER CHECK. AFMS

1. **AIR SOURCE HEAT/START** switch - **ENG**.
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS** switch - As required for tanks installed.
4. **XFER FROM** switch - **OUTBD**.
5. **MAN XFER** switch - **BOTH**.
6. **XFER MODE** switch - **MAN**.
7. **UH-60A TOTAL FUEL** or **UH-60L MAIN FUEL** readout - Check for increase of approximately 40 pounds and **AUX FUEL QTY LBS** decrease of 20 pounds per tank.
8. **XFER FROM** switch - **INBD** if installed.
9. **UH-60A TOTAL FUEL** or **UH-60L MAIN FUEL** readout - Check for increase of approximately 40 pounds and **AUX FUEL QTY LBS** decrease of 20 pounds per tank.
10. **XFER MODE** switch - **OFF**.

11. External ERFs - Set as desired.

FUEL TRANSFER IN AUTO MODE. AFMS

1. **AIR SOURCE HEAT/START** switch - **ENG**.
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS** switch - As required for tanks installed.
4. **XFER FROM** switch - **OUTBD**, then **INBD**.
5. **XFER MODE** switch - **AUTO**.

FUEL TRANSFER IN MANUAL MODE. AFMS

If the **AUTO** mode is inoperative or a lateral imbalance greater than 400 pounds between tank pairs is identified, transfer in the **MAN** mode as follows:

1. **AIR SOURCE HEAT/START** switch - **ENG**.
2. **FUEL BOOST PUMP CONTROL** switches - Check **ON**.
3. **PRESS** switch - As required for tanks installed.
4. **XFER FROM** switch - **OUTBD**, then **INBD**.
5. **MAN XFER** switch - **BOTH** or select heavy tank to correct an imbalance.
6. **XFER MODE** switch - **MAN**.

EXTERNAL AUXILIARY FUEL MANAGEMENT SYSTEM FUEL TRANSFER CHECK. **CEFS**

1. **XFER FROM** switch - **OUTBD**.
2. **MAN XFER** switch - **BOTH**.
3. **XFER MODE** switch - **MAN**.
4. **UH-60A** **TOTAL FUEL** or **UH-60L** **MAIN FUEL** readout - Check for increase of approximately 40 pounds and **AUX FUEL QTY LBS** decrease of 20 pounds per tank.
5. **XFER FROM** switch - **INBD** if installed.
6. **UH-60A** **TOTAL FUEL** or **UH-60L** **MAIN FUEL** readout - Check for increase of approximately 40 pounds and **AUX FUEL QTY LBS** decrease of 20 pounds per tank.
7. **XFER MODE** switch - **OFF**.

FUEL TRANSFER IN AUTO MODE. **CEFS**

1. **XFER FROM** switch - **OUTBD**, then **INBD**.
2. **XFER MODE** switch - **AUTO**.

FUEL TRANSFER IN MANUAL MODE. **CEFS**

If the **AUTO** mode is inoperative or a lateral imbalance (270 pounds for outboard tanks or 375 pounds for inboard tanks) is identified, transfer in the **MAN** mode as follows:

1. **XFER FROM** switch - **OUTBD**, then **INBD**.

2. **MAN XFER** switch - **BOTH** or select heavy tank to correct an imbalance.
3. **XFER MODE** switch - **MAN**.

VOLCANO MULTIPLE MINE DELIVERY SYSTEM.

MINE LAUNCH.

1. **DCU FIRE CIRCUIT** switch safety pin and streamer - Remove.
2. **DCU FIRE CIRCUIT** switch - **ENABLE**.
3. Before reaching target, **VOLCANO ARM** switch - **ARM**. Verify **P/F/ARMED** indicates **ARMED**.
4. Verify that the **HELICOPTER DELIVERY SPEED** settings agree with the helicopter ground speed.
5. **GA** button - Press either pilot's to start launching mines. Press either **GA** button a second time to stop mine launching.
6. If launch is interrupted longer than 60 seconds, resume launch: **VOLCANO ARM** switch - Off for at least sixteen seconds.
7. **VOLCANO ARM** switch - **ARM**. Verify a steady **ARMED** is displayed on the ICP. Launching can then be resumed.
8. During mine launching, if an error code appears on DCU panel that affects mission performance, do the following:
 - a. **DCU FIRE CIRCUIT** switch - **OFF**.

- b. Safety pin and streamer assembly - Install to **FIRE CIRCUIT** switch.
- c. DCU **POWER** switch - **OFF**.
- d. Return to downloading area and remove canisters, refer to TM 9-1095-208-23-2&P.

POST MINE LAUNCH CHECK.

- 1. ICP **VOLCANO ARM** switch - Off (down).
- 2. DCU **FIRE CIRCUIT** switch - **OFF**.
- 3. Safety pin and streamer assembly - Install to **FIRE CIRCUIT** switch.
- 4. DCU **POWER** switch - **OFF**.

CREW, PASSENGERS, AND MISSION EQUIPMENT - SECURE.

- a. **EH** ASE - Check as required.
- O b. **EH** ECM ANTENNA switch - **RETRACT**.
Check **ANTENNA RETRACTED** advisory - Appears. ECM operator report antenna deployed light- Off.
- O c. **EH** IINS TACAN - **OFF**.

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE**

HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-50	-45	-40	-35	-30	-25	-20
0	1.0	114	114	115	114	114	116	118
	0.9	109	109	109	109	109	111	113
500	1.0	112	112	113	112	112	114	116
	0.9	107	107	107	107	107	109	110
1000	1.0	110	110	111	111	111	112	113
	0.9	105	105	105	106	106	107	108
1500	1.0	108	108	109	109	109	110	111
	0.9	103	103	104	104	104	105	106
2000	1.0	106	106	107	107	107	108	109
	0.9	101	102	102	102	102	103	104
2500	1.0	104	105	105	105	105	106	107
	0.9	100	100	100	100	100	101	102
3000	1.0	103	103	103	103	103	104	105
	0.9	98	98	98	98	98	99	100
3500	1.0	101	101	101	101	101	102	103
	0.9	96	96	96	96	97	97	99
4000	1.0	99	99	99	99	99	100	101
	0.9	94	94	94	95	95	96	97
4500	1.0	97	97	97	97	98	98	100
	0.9	93	93	93	93	93	94	95
5000	1.0	95	95	95	96	96	97	98
	0.9	91	91	91	91	91	92	93
5500	1.0	93	94	94	94	94	95	96
	0.9	89	89	89	90	90	91	92
6000	1.0	92	92	92	92	92	93	94
	0.9	88	88	88	88	88	89	90
6500	1.0	90	90	90	90	91	91	92
	0.9	86	86	86	86	87	87	88
7000	1.0	88	88	89	89	89	90	91
	0.9	84	84	85	85	85	86	87
7500	1.0	87	87	87	87	87	88	89
	0.9	83	83	83	83	83	84	85

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 100%

AA9971_1A_CL
SAF

Figure 1. Maximum Torque Available (Sheet 1 of 6) 700

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-50	-45	-40	-35	-30	-25	-20
8000	1.0	85	85	85	85	86	86	87
	0.9	81	81	81	82	82	82	83
8500	1.0	84	84	84	84	84	85	86
	0.9	80	80	80	80	80	81	82
9000	1.0	82	82	82	82	83	83	84
	0.9	78	78	78	79	79	80	80
9500	1.0	80	80	81	81	81	82	83
	0.9	77	77	77	77	77	78	79
10,000	1.0	79	79	79	79	80	80	81
	0.9	75	75	75	76	76	77	77
10,500	1.0	77	77	77	78	78	79	79
	0.9	74	74	74	74	74	75	76
11,000	1.0	76	76	76	76	76	77	78
	0.9	72	73	73	73	73	74	74
11,500	1.0	74	75	75	75	75	76	76
	0.9	71	71	71	71	71	72	73
12,000	1.0	73	73	73	73	73	74	75
	0.9	70	70	70	70	70	71	72
12,500	1.0	72	72	72	72	72	73	74
	0.9	68	68	68	68	69	69	70
13,000	1.0	70	70	70	70	70	71	72
	0.9	67	67	67	67	67	68	69
13,500	1.0	69	69	69	69	69	70	71
	0.9	66	66	66	66	66	67	67
14,000	1.0	67	67	67	67	68	68	69
	0.9	64	64	64	64	65	65	66
14,500	1.0	66	66	66	66	66	67	68
	0.9	63	63	63	63	63	64	65
15,000	1.0	65	65	65	65	65	66	67
	0.9	62	62	62	62	62	63	64
15,500	1.0	63	63	63	63	64	64	65
	0.9	61	61	60	61	61	62	62

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 100%

AA9971_2A_CL
SAF

Figure 1. Maximum Torque Available (Sheet 2 of 6) 700

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-15	-10	-5	0	5	10	15
0	1.0	117	117	117	116	116	115	113
	0.9	112	112	111	110	109	108	105
500	1.0	116	116	116	115	114	113	111
	0.9	110	110	111	109	108	107	103
1000	1.0	114	114	116	115	113	111	109
	0.9	109	109	110	109	107	105	102
1500	1.0	112	113	115	114	112	110	107
	0.9	107	108	110	108	106	103	100
2000	1.0	110	112	115	113	111	108	106
	0.9	105	107	109	107	104	102	98
2500	1.0	108	110	113	111	109	106	104
	0.9	103	105	107	105	103	100	97
3000	1.0	106	108	111	109	107	104	102
	0.9	102	103	106	104	101	98	95
3500	1.0	104	106	109	107	105	102	100
	0.9	100	101	104	102	99	97	93
4000	1.0	103	104	107	105	103	101	99
	0.9	98	99	102	100	97	95	92
4500	1.0	101	102	105	103	101	99	97
	0.9	96	98	100	98	95	93	90
5000	1.0	99	101	103	101	99	97	95
	0.9	95	96	98	96	93	91	88
5500	1.0	97	99	101	99	97	95	93
	0.9	93	94	96	94	91	89	86
6000	1.0	95	97	99	97	95	93	91
	0.9	91	93	94	92	90	87	84
6500	1.0	94	95	97	95	93	91	89
	0.9	89	91	93	90	88	86	83
7000	1.0	92	94	95	93	91	89	87
	0.9	88	89	91	88	86	84	81
7500	1.0	90	92	94	91	89	87	85
	0.9	86	88	89	87	84	82	79

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 100%

AA9971_3A_CL
SAF

Figure 1. Maximum Torque Available (Sheet 3 of 6) 700

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-15	-10	-5	0	5	10	15
8000	1.0	89	90	92	90	88	85	84
	0.9	85	86	88	85	83	81	78
8500	1.0	87	89	90	88	86	84	82
	0.9	83	85	86	83	81	79	76
9000	1.0	85	87	88	86	84	82	80
	0.9	82	83	84	82	79	77	75
9500	1.0	84	85	86	84	82	80	79
	0.9	80	81	82	80	78	76	73
10,000	1.0	82	84	85	83	81	79	77
	0.9	78	80	81	78	76	74	72
10,500	1.0	81	82	83	81	79	77	75
	0.9	77	78	79	77	75	73	70
11,000	1.0	79	80	81	79	77	75	74
	0.9	75	77	77	75	73	71	69
11,500	1.0	78	79	79	78	76	74	72
	0.9	74	75	76	74	71	70	67
12,000	1.0	76	77	78	76	74	72	71
	0.9	73	74	74	72	70	68	66
12,500	1.0	75	76	76	74	72	71	69
	0.9	71	72	73	70	68	67	64
13,000	1.0	73	74	74	73	71	69	67
	0.9	70	71	71	69	67	65	63
13,500	1.0	72	73	73	71	69	68	66
	0.9	68	70	69	67	66	64	61
14,000	1.0	70	71	71	70	68	66	64
	0.9	67	68	68	66	64	62	60
14,500	1.0	69	70	70	68	66	65	63
	0.9	66	67	66	64	63	61	59
15,000	1.0	68	69	68	66	65	63	62
	0.9	65	66	65	63	61	60	57
15,500	1.0	66	67	67	65	63	62	60
	0.9	63	64	64	62	60	58	56

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 100%

AA9971_4A_CL
SAF

Figure 1. Maximum Torque Available (Sheet 4 of 6) 700

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		20	25	30	35	40	45	50
0	1.0	111	108	105	100	95	91	86
	0.9	102	98	94	90	86	82	77
500	1.0	109	106	103	98	94	89	84
	0.9	100	96	93	88	84	80	76
1000	1.0	107	104	101	97	92	87	83
	0.9	98	95	91	87	83	79	74
1500	1.0	105	102	99	95	91	86	81
	0.9	97	93	89	85	81	77	73
2000	1.0	103	100	97	93	89	84	80
	0.9	95	91	88	84	80	76	72
2500	1.0	102	99	96	91	87	83	78
	0.9	93	90	86	82	78	74	70
3000	1.0	100	97	94	90	86	81	77
	0.9	92	88	85	81	77	73	69
3500	1.0	98	95	92	88	84	80	75
	0.9	90	87	83	79	76	72	68
4000	1.0	96	93	91	86	82	78	74
	0.9	88	85	82	78	74	70	66
4500	1.0	94	91	89	85	81	76	72
	0.9	87	83	80	76	72	69	65
5000	1.0	92	90	87	83	79	75	70
	0.9	85	81	78	74	71	67	63
5500	1.0	90	88	85	81	77	73	69
	0.9	83	80	77	73	69	66	62
6000	1.0	88	86	83	79	75	71	67
	0.9	81	78	75	71	68	64	60
6500	1.0	87	84	81	77	74	70	66
	0.9	80	76	73	70	66	63	59
7000	1.0	85	82	80	76	72	68	64
	0.9	78	75	72	68	65	61	58
7500	1.0	83	81	78	74	70	66	63
	0.9	77	73	70	67	63	60	56

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%

■ AREA IS ABOVE ENGINE AMBIENT TEMPERATURE LIMIT

AA9971_5A_CL
SAF

Figure 1. Maximum Torque Available (Sheet 5 of 6) 700

**MAXIMUM TORQUE AVAILABLE* – 30 MINUTE LIMIT
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

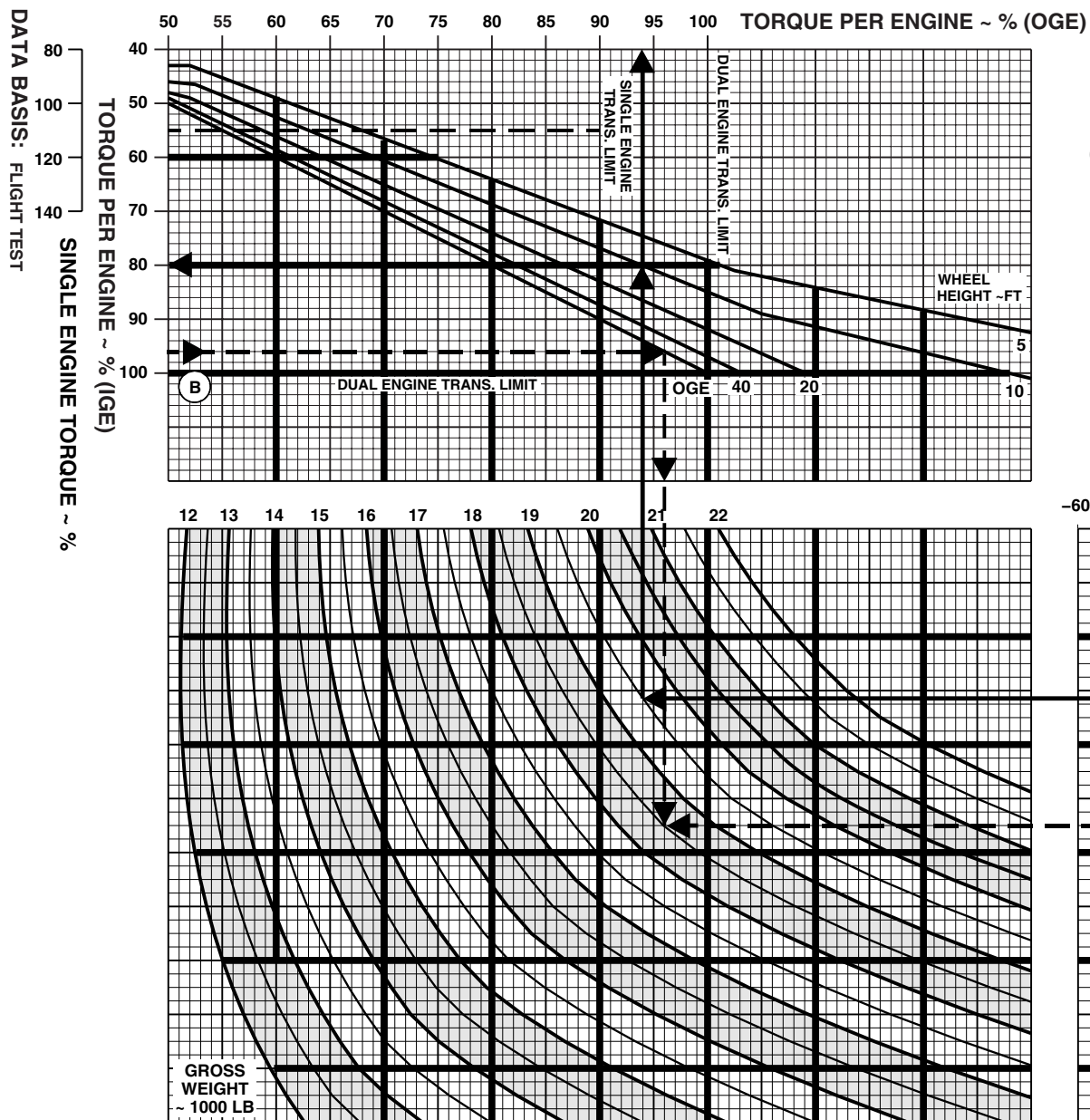
HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		20	25	30	35	40	45	50
8000	1.0	82	79	76	73	69	65	61
	0.9	75	72	69	65	62	58	55
8500	1.0	80	77	75	71	67	63	60
	0.9	73	70	67	64	61	57	54
9000	1.0	78	76	73	69	66	62	58
	0.9	72	69	66	62	59	56	52
9500	1.0	77	74	71	68	64	61	57
	0.9	70	67	64	61	58	54	51
10,000	1.0	75	72	70	66	63	59	56
	0.9	69	66	63	60	56	53	50
10,500	1.0	74	71	68	65	61	58	54
	0.9	67	64	62	58	55	52	49
11,000	1.0	72	69	67	63	60	56	53
	0.9	66	63	60	57	54	51	48
11,500	1.0	70	68	65	62	58	55	51
	0.9	65	62	59	55	52	49	46
12,000	1.0	69	66	63	60	57	53	50
	0.9	63	60	57	54	51	48	45
12,500	1.0	67	65	62	59	55	52	49
	0.9	62	59	56	53	50	47	44
13,000	1.0	66	63	60	57	54	51	48
	0.9	60	57	55	51	49	46	43
13,500	1.0	64	62	59	56	53	49	46
	0.9	59	56	53	50	47	45	42
14,000	1.0	63	60	57	54	51	48	45
	0.9	58	55	52	49	46	43	41
14,500	1.0	61	59	56	53	50	47	44
	0.9	56	53	51	48	45	42	40
15,000	1.0	60	57	55	52	49	46	43
	0.9	55	52	49	46	44	41	39
15,500	1.0	58	56	53	50	47	45	42
	0.9	54	51	48	45	43	40	37

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 100%

AREA IS ABOVE ENGINE AMBIENT TEMPERATURE LIMIT

AA9971_6_CL
SAF

Figure 1. Maximum Torque Available (Sheet 6 of 6) 700



HOVER
CLEAN
T700(2)

HOVER CLEAN CONFIGURATION 100% RPM R ZERO WIND

NOTE

FOR LOW WIND CONDITIONS
AIRCRAFT SHOULD BE HEADED
INTO WIND. 3-5 KT CROSSWIND
OR TAILWIND MAY INCREASE
TORQUE REQUIRED BY UP TO
4% OVER ZERO WIND VALUES

FREE AIR TEMP. ~ °C

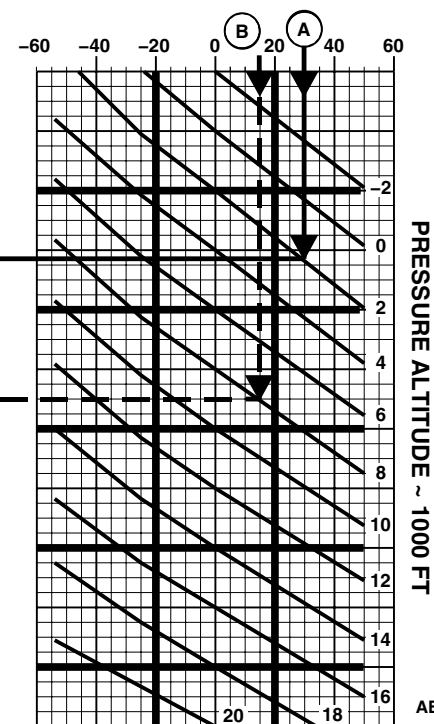
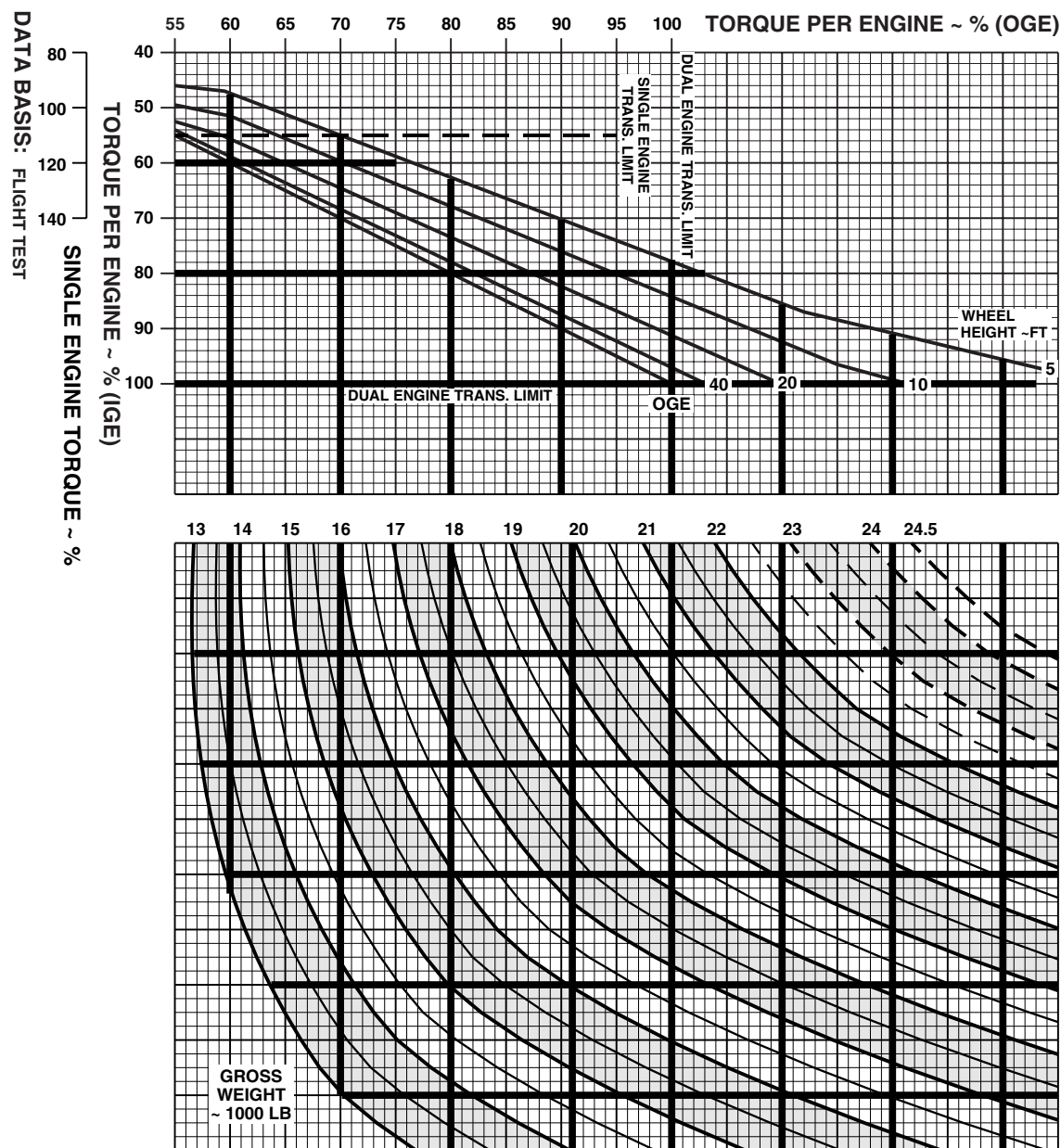


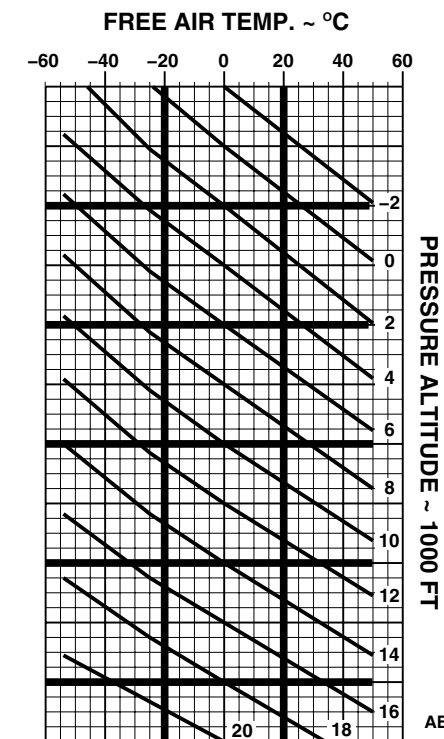
Figure 2. Hover - Clean Configuration **700**



HOVER **HIGH DRAG CONFIGURATION** **100% RPM R** **ZERO WIND**

NOTE

FOR LOW WIND CONDITIONS
 AIRCRAFT SHOULD BE HEADED
 INTO WIND. 3-5 KT CROSSWIND
 OR TAILWIND MAY INCREASE
 TORQUE REQUIRED BY UP TO
 4% OVER ZERO WIND VALUES



AB2494CL
 SA

Figure 3. Hover - High Drag Configuration **700**

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED ANTI-ICE OFF T700-GE-700 ENGINE

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	98	98	99	98	100	99
	Q ~ IGE ~ %	85	85	86	85	86	86
1000	GW ~ 100 LB	220	220	219	219	218	218
	Q ~ OGE ~ %	99	99	100	100	100	100
	Q ~ IGE ~ %	86	86	86	86	86	86
2000	GW ~ 100 LB	219	218	217	216	215	215
	Q ~ OGE ~ %	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	85	86	85
2500	GW ~ 100 LB	218	216	215	215	214	214
	Q ~ OGE ~ %	100	99	100	99	100	100
	Q ~ IGE ~ %	86	85	86	85	86	85
3000	GW ~ 100 LB	216	212	214	211	213	210
	Q ~ OGE ~ %	100	98	100	98	100	98
	Q ~ IGE ~ %	86	83	86	83	86	83
3500	GW ~ 100 LB	215	208	213	207	211	206
	Q ~ OGE ~ %	100	96	100	96	100	96
	Q ~ IGE ~ %	86	82	85	82	85	82
4000	GW ~ 100 LB	212	204	210	203	209	202
	Q ~ OGE ~ %	99	94	99	94	99	94
	Q ~ IGE ~ %	84	80	84	80	84	80
4500	GW ~ 100 LB	208	201	206	200	205	199
	Q ~ OGE ~ %	97	92	97	92	97	92
	Q ~ IGE ~ %	83	79	83	79	83	79
5000	GW ~ 100 LB	204	197	202	196	201	195
	Q ~ OGE ~ %	95	91	95	91	95	91
	Q ~ IGE ~ %	81	77	81	77	81	77
5500	GW ~ 100 LB	200	193	198	192	198	192
	Q ~ OGE ~ %	94	89	94	89	94	89
	Q ~ IGE ~ %	80	76	80	76	80	76

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AB2925_1_CL
SAF

Figure 4. Hover Torque Required (Sheet 1 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
6000	GW ~ 100 LB	196	190	195	189	194	188
	Q ~ OGE ~ %	92	87	92	87	92	87
	Q ~ IGE ~ %	78	74	78	74	78	74
6500	GW ~ 100 LB	193	186	191	185	190	185
	Q ~ OGE ~ %	90	86	90	86	90	86
	Q ~ IGE ~ %	77	73	77	73	77	73
7000	GW ~ 100 LB	189	183	187	182	187	181
	Q ~ OGE ~ %	88	84	89	84	89	84
	Q ~ IGE ~ %	76	72	76	72	76	72
7500	GW ~ 100 LB	186	179	184	178	183	178
	Q ~ OGE ~ %	87	82	87	82	87	83
	Q ~ IGE ~ %	74	70	74	70	74	70
8000	GW ~ 100 LB	182	176	180	175	180	174
	Q ~ OGE ~ %	85	81	85	81	86	81
	Q ~ IGE ~ %	73	69	73	69	73	69
8500	GW ~ 100 LB	179	172	177	172	177	171
	Q ~ OGE ~ %	84	79	84	79	84	80
	Q ~ IGE ~ %	71	68	71	68	72	68
9000	GW ~ 100 LB	175	169	174	168	173	168
	Q ~ OGE ~ %	82	78	82	78	82	78
	Q ~ IGE ~ %	70	66	70	66	70	67
9500	GW ~ 100 LB	172	166	170	165	170	165
	Q ~ OGE ~ %	80	76	80	76	81	77
	Q ~ IGE ~ %	69	65	69	65	69	65
10,000	GW ~ 100 LB	169	163	167	162	167	162
	Q ~ OGE ~ %	79	75	79	75	79	75
	Q ~ IGE ~ %	67	64	67	64	67	64
10,500	GW ~ 100 LB	166	160	164	159	163	158
	Q ~ OGE ~ %	77	73	78	74	78	74
	Q ~ IGE ~ %	66	63	66	63	66	63

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AB2925_2_CL
SAF

Figure 4. Hover Torque Required (Sheet 2 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
11,000	GW ~ 100 LB	162	157	161	156	160	155
	Q ~ OGE ~ %	76	72	76	72	76	72
	Q ~ IGE ~ %	65	61	65	61	65	62
11,500	GW ~ 100 LB	159	154	158	153	157	152
	Q ~ OGE ~ %	75	71	75	71	75	71
	Q ~ IGE ~ %	64	60	64	60	64	60
12,000	GW ~ 100 LB	156	151	155	150	154	149
	Q ~ OGE ~ %	73	69	73	69	73	69
	Q ~ IGE ~ %	62	59	62	59	62	59
12,500	GW ~ 100 LB	153	148	152	147	151	146
	Q ~ OGE ~ %	72	68	72	68	72	68
	Q ~ IGE ~ %	61	58	61	58	61	58
13,000	GW ~ 100 LB	150	145	148	144	148	143
	Q ~ OGE ~ %	70	67	70	67	70	67
	Q ~ IGE ~ %	60	57	60	57	60	57
13,500	GW ~ 100 LB	147	142	145	141	145	140
	Q ~ OGE ~ %	69	65	69	65	69	65
	Q ~ IGE ~ %	59	56	59	56	59	56
14,000	GW ~ 100 LB	144	139	142	138	142	138
	Q ~ OGE ~ %	68	64	67	64	67	64
	Q ~ IGE ~ %	58	54	57	54	57	54
14,500	GW ~ 100 LB	141	136	140	135	139	135
	Q ~ OGE ~ %	66	63	66	63	66	63
	Q ~ IGE ~ %	56	53	56	53	56	53
15,000	GW ~ 100 LB	139	133	137	133	136	132
	Q ~ OGE ~ %	65	61	65	61	65	61
	Q ~ IGE ~ %	55	52	55	52	55	52
15,500	GW ~ 100 LB	136	131	134	130	134	129
	Q ~ OGE ~ %	64	60	63	60	63	60
	Q ~ IGE ~ %	54	51	54	51	54	51

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AB2925_3_CL
SAF

Figure 4. Hover Torque Required (Sheet 3 of 15) 700

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	219	219	218	218	217	217	216	216
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	86	86	86	86	86	86	86
1000	GW ~ 100 LB	217	217	216	216	215	215	214	214
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	85	86	85	86	85
2000	GW ~ 100 LB	214	214	213	213	212	212	211	211
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	85	86	85	86	85
2500	GW ~ 100 LB	213	213	212	212	211	211	210	210
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	85	86	85	86	85
3000	GW ~ 100 LB	212	209	211	210	210	210	209	209
	Q ~ OGE ~ %	100	98	100	99	100	100	100	100
	Q ~ IGE ~ %	86	83	85	84	85	84	85	84
3500	GW ~ 100 LB	210	206	209	206	208	206	207	207
	Q ~ OGE ~ %	100	96	100	97	100	98	100	99
	Q ~ IGE ~ %	85	82	85	82	85	83	85	84
4000	GW ~ 100 LB	208	202	208	202	207	202	206	203
	Q ~ OGE ~ %	99	94	100	95	100	96	100	98
	Q ~ IGE ~ %	84	80	85	81	85	82	85	83
4500	GW ~ 100 LB	204	198	205	198	205	199	205	200
	Q ~ OGE ~ %	97	93	99	94	99	95	100	96
	Q ~ IGE ~ %	83	79	84	79	84	80	85	81
5000	GW ~ 100 LB	201	195	201	195	201	195	202	196
	Q ~ OGE ~ %	96	91	97	92	98	93	99	94
	Q ~ IGE ~ %	82	77	82	78	83	79	84	80
5500	GW ~ 100 LB	197	191	197	191	198	192	199	193
	Q ~ OGE ~ %	94	89	95	90	96	91	97	93
	Q ~ IGE ~ %	80	76	81	77	82	77	83	78

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE ≡ 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AA9977_1A_CL
SAF

Figure 4. Hover Torque Required (Sheet 4 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
6000	GW ~ 100 LB	193	187	194	188	194	188	195	189
	Q ~ OGE ~ %	92	88	93	89	94	89	96	91
	Q ~ IGE ~ %	79	75	79	75	80	76	81	77
6500	GW ~ 100 LB	190	184	190	184	190	185	191	186
	Q ~ OGE ~ %	91	86	92	87	92	88	94	89
	Q ~ IGE ~ %	77	73	78	74	79	74	80	75
7000	GW ~ 100 LB	186	181	186	181	187	181	188	182
	Q ~ OGE ~ %	89	85	90	85	91	86	92	88
	Q ~ IGE ~ %	76	72	76	72	77	73	78	74
7500	GW ~ 100 LB	183	177	183	177	183	178	184	179
	Q ~ OGE ~ %	87	83	88	84	89	85	90	86
	Q ~ IGE ~ %	74	71	75	71	76	72	77	73
8000	GW ~ 100 LB	179	174	179	174	180	174	181	175
	Q ~ OGE ~ %	86	81	86	82	87	83	89	84
	Q ~ IGE ~ %	73	69	74	70	74	70	75	71
8500	GW ~ 100 LB	176	171	176	171	176	171	177	172
	Q ~ OGE ~ %	84	80	85	81	86	81	87	83
	Q ~ IGE ~ %	72	68	72	68	73	69	74	70
9000	GW ~ 100 LB	173	167	173	168	173	168	174	169
	Q ~ OGE ~ %	83	78	83	79	84	80	86	81
	Q ~ IGE ~ %	70	67	71	67	72	68	73	69
9500	GW ~ 100 LB	169	164	170	164	170	165	171	166
	Q ~ OGE ~ %	81	77	82	78	83	78	84	80
	Q ~ IGE ~ %	69	65	70	66	70	66	71	67
10,000	GW ~ 100 LB	166	161	166	161	167	161	167	162
	Q ~ OGE ~ %	79	75	80	76	81	77	82	78
	Q ~ IGE ~ %	67	64	68	65	69	65	70	66
10,500	GW ~ 100 LB	163	158	163	158	163	158	164	159
	Q ~ OGE ~ %	78	74	79	75	79	75	81	77
	Q ~ IGE ~ %	66	63	67	63	68	64	69	65

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE ≡ 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AA9977_2A_CL
SAF

Figure 4. Hover Torque Required (Sheet 5 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
11,000	GW ~ 100 LB	160	155	160	155	160	155	161	156
	Q ~ OGE ~ %	76	72	77	73	78	74	79	75
	Q ~ IGE ~ %	65	62	66	62	66	63	67	63
11,500	GW ~ 100 LB	157	152	157	152	157	152	158	153
	Q ~ OGE ~ %	75	71	76	72	76	73	78	74
	Q ~ IGE ~ %	64	60	64	61	65	61	66	62
12,000	GW ~ 100 LB	153	149	154	149	154	149	155	150
	Q ~ OGE ~ %	73	70	74	70	75	71	76	72
	Q ~ IGE ~ %	62	59	63	60	64	60	65	61
12,500	GW ~ 100 LB	150	146	151	146	151	147	152	147
	Q ~ OGE ~ %	72	68	73	69	74	70	75	71
	Q ~ IGE ~ %	61	58	62	58	62	59	63	60
13,000	GW ~ 100 LB	147	143	148	143	148	144	149	144
	Q ~ OGE ~ %	70	67	71	68	72	68	73	69
	Q ~ IGE ~ %	60	57	61	57	61	58	62	59
13,500	GW ~ 100 LB	144	140	145	140	145	141	146	142
	Q ~ OGE ~ %	69	65	70	66	71	67	72	68
	Q ~ IGE ~ %	59	56	59	56	60	57	61	58
14,000	GW ~ 100 LB	141	137	142	138	142	138	143	139
	Q ~ OGE ~ %	68	64	68	65	69	66	70	67
	Q ~ IGE ~ %	57	54	58	55	59	56	60	56
14,500	GW ~ 100 LB	139	134	139	135	140	135	140	136
	Q ~ OGE ~ %	66	63	67	64	68	64	69	65
	Q ~ IGE ~ %	56	53	57	54	58	55	59	55
15,000	GW ~ 100 LB	136	132	136	132	137	133	138	134
	Q ~ OGE ~ %	65	62	66	62	67	63	68	64
	Q ~ IGE ~ %	55	52	56	53	57	54	57	54
15,500	GW ~ 100 LB	133	129	134	130	134	130	135	131
	Q ~ OGE ~ %	64	60	65	61	65	62	66	63
	Q ~ IGE ~ %	54	51	55	52	56	52	56	53

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE ≡ 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AA9977_3A_CL
SAF

Figure 4. Hover Torque Required (Sheet 6 of 15) 700

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	215	215	214	214	214	214	213	213
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	86	86	86	86	86	86	86
1000	GW ~ 100 LB	213	213	212	212	211	211	210	210
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	86	86	86	86	86
2000	GW ~ 100 LB	210	210	210	210	209	209	208	208
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	86	85	86	86	86	86	86	86
3000	GW ~ 100 LB	208	208	207	207	206	206	205	205
	Q ~ OGE ~ %	100	100	100	100	100	100	100	100
	Q ~ IGE ~ %	85	84	85	85	85	85	85	85
4000	GW ~ 100 LB	205	204	204	204	203	203	203	199
	Q ~ OGE ~ %	100	99	100	100	100	100	100	97
	Q ~ IGE ~ %	85	84	85	85	85	85	85	83
4500	GW ~ 100 LB	204	201	203	200	202	199	201	195
	Q ~ OGE ~ %	100	97	100	98	100	98	100	95
	Q ~ IGE ~ %	85	82	85	83	85	83	85	81
5000	GW ~ 100 LB	202	197	202	197	201	195	198	191
	Q ~ OGE ~ %	100	96	100	96	100	96	99	93
	Q ~ IGE ~ %	85	81	85	82	85	82	84	80
5500	GW ~ 100 LB	200	194	199	193	198	191	194	187
	Q ~ OGE ~ %	99	94	99	94	99	94	97	91
	Q ~ IGE ~ %	84	79	84	80	84	80	82	78
6000	GW ~ 100 LB	196	190	195	189	194	188	190	184
	Q ~ OGE ~ %	97	92	97	93	97	92	95	90
	Q ~ IGE ~ %	82	78	82	79	82	78	80	76
6500	GW ~ 100 LB	192	187	191	186	190	184	187	180
	Q ~ OGE ~ %	95	91	95	91	95	90	93	88
	Q ~ IGE ~ %	81	76	81	77	81	77	79	75

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE ≡ 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9978_1A_CL
SAF

Figure 4. Hover Torque Required (Sheet 7 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
7000	GW ~ 100 LB	189	183	188	182	187	181	183	177
	Q ~ OGE ~ %	94	89	94	89	93	88	91	86
	Q ~ IGE ~ %	79	75	79	76	79	75	77	73
7500	GW ~ 100 LB	185	180	184	179	183	177	180	173
	Q ~ OGE ~ %	92	87	92	88	92	87	89	84
	Q ~ IGE ~ %	78	74	78	75	78	74	76	72
8000	GW ~ 100 LB	182	177	181	175	180	174	176	170
	Q ~ OGE ~ %	90	86	90	86	90	85	87	83
	Q ~ IGE ~ %	76	72	76	73	76	72	74	71
8500	GW ~ 100 LB	178	173	177	172	176	170	173	166
	Q ~ OGE ~ %	88	84	88	84	88	83	86	81
	Q ~ IGE ~ %	75	71	75	72	75	71	73	69
9000	GW ~ 100 LB	175	170	174	169	173	167	169	163
	Q ~ OGE ~ %	87	82	86	83	86	82	84	79
	Q ~ IGE ~ %	74	70	73	70	73	70	71	68
9500	GW ~ 100 LB	172	167	170	165	169	164	166	160
	Q ~ OGE ~ %	85	81	85	81	84	80	82	78
	Q ~ IGE ~ %	72	68	72	69	72	68	70	66
10,000	GW ~ 100 LB	168	163	167	162	166	160	163	157
	Q ~ OGE ~ %	84	79	83	79	83	78	81	76
	Q ~ IGE ~ %	71	67	70	68	70	67	69	65
10,500	GW ~ 100 LB	165	160	164	159	162	157	159	153
	Q ~ OGE ~ %	82	78	81	78	81	77	79	75
	Q ~ IGE ~ %	69	66	69	66	69	65	67	64
11,000	GW ~ 100 LB	162	157	161	156	159	154	156	150
	Q ~ OGE ~ %	80	76	80	76	79	75	77	73
	Q ~ IGE ~ %	68	64	68	65	67	64	66	62
11,500	GW ~ 100 LB	159	154	157	153	156	151	153	147
	Q ~ OGE ~ %	79	75	78	75	78	74	76	72
	Q ~ IGE ~ %	67	63	66	64	66	63	64	61

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE ≡ 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9978_2A_CL
SAF

Figure 4. Hover Torque Required (Sheet 8 of 15) 700

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
12,000	GW ~ 100 LB	156	151	154	150	153	147	150	144
	Q ~ OGE ~ %	77	73	76	73	76	72	74	70
	Q ~ IGE ~ %	65	62	65	62	64	61	63	60
12,500	GW ~ 100 LB	153	148	151	147	149	144	147	141
	Q ~ OGE ~ %	76	72	75	72	74	70	73	69
	Q ~ IGE ~ %	64	61	64	61	63	60	62	59
13,000	GW ~ 100 LB	150	145	148	144	146	141	143	138
	Q ~ OGE ~ %	74	70	73	70	73	69	71	67
	Q ~ IGE ~ %	63	59	62	60	62	59	60	57
13,500	GW ~ 100 LB	147	142	145	141	143	138	140	135
	Q ~ OGE ~ %	73	69	72	69	71	67	69	66
	Q ~ IGE ~ %	62	58	61	59	60	58	59	56
14,000	GW ~ 100 LB	144	140	142	138	140	135	137	132
	Q ~ OGE ~ %	71	68	70	67	70	66	68	64
	Q ~ IGE ~ %	60	57	60	57	59	56	58	55
14,500	GW ~ 100 LB	141	137	139	135	137	133	134	130
	Q ~ OGE ~ %	70	66	69	66	68	65	66	63
	Q ~ IGE ~ %	59	56	58	56	58	55	56	54
15,000	GW ~ 100 LB	138	134	136	132	134	130	132	127
	Q ~ OGE ~ %	69	65	68	64	66	63	65	61
	Q ~ IGE ~ %	58	55	57	55	56	54	55	52
15,500	GW ~ 100 LB	136	132	133	130	131	127	129	124
	Q ~ OGE ~ %	67	64	66	63	65	62	63	60
	Q ~ IGE ~ %	57	54	56	54	55	53	54	51

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9978_3A_CL
SAF

Figure 4. Hover Torque Required (Sheet 9 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	212	212	211	211	210	210	209	206
	Q ~ OGE ~ %	100	100	100	100	100	100	100	98
	Q ~ IGE ~ %	86	86	86	86	86	86	86	84
1000	GW ~ 100 LB	209	209	208	208	207	205	206	199
	Q ~ OGE ~ %	100	100	100	100	100	98	100	95
	Q ~ IGE ~ %	86	86	85	86	86	84	86	81
2000	GW ~ 100 LB	207	207	206	204	205	198	204	192
	Q ~ OGE ~ %	100	100	100	98	100	95	100	91
	Q ~ IGE ~ %	86	86	85	84	85	81	85	79
2500	GW ~ 100 LB	205	205	204	200	203	194	201	188
	Q ~ OGE ~ %	100	100	100	97	100	93	99	90
	Q ~ IGE ~ %	85	85	85	83	85	80	84	77
3000	GW ~ 100 LB	204	202	203	196	202	191	197	185
	Q ~ OGE ~ %	100	98	100	95	100	92	97	88
	Q ~ IGE ~ %	85	84	85	81	85	79	83	76
3500	GW ~ 100 LB	203	198	201	193	198	187	194	182
	Q ~ OGE ~ %	100	97	100	93	98	90	95	87
	Q ~ IGE ~ %	85	82	95	80	84	77	81	74
4000	GW ~ 100 LB	201	194	198	189	195	184	190	178
	Q ~ OGE ~ %	100	95	98	92	96	88	93	85
	Q ~ IGE ~ %	85	80	84	78	82	76	80	73
4500	GW ~ 100 LB	198	191	195	186	191	180	186	175
	Q ~ OGE ~ %	99	93	96	90	94	87	91	83
	Q ~ IGE ~ %	84	79	82	77	80	74	78	72
5000	GW ~ 100 LB	195	187	191	182	187	177	183	171
	Q ~ OGE ~ %	97	91	94	88	92	85	90	81
	Q ~ IGE ~ %	82	77	80	75	79	72	76	70
5500	GW ~ 100 LB	191	184	187	179	184	173	179	168
	Q ~ OGE ~ %	95	89	92	86	90	83	88	80
	Q ~ IGE ~ %	80	76	79	74	77	71	75	69

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9979_1_CL
SAF

Figure 4. Hover Torque Required (Sheet 10 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
6000	GW ~ 100 LB	187	180	184	175	180	170	176	165
	Q ~ OGE ~ %	93	87	91	84	88	81	86	78
	Q ~ IGE ~ %	79	75	77	72	75	70	73	67
6500	GW ~ 100 LB	183	177	180	172	177	167	172	161
	Q ~ OGE ~ %	91	86	89	83	87	80	84	76
	Q ~ IGE ~ %	77	73	76	71	74	68	72	66
7000	GW ~ 100 LB	181	173	176	168	173	163	169	158
	Q ~ OGE ~ %	89	84	87	81	85	78	82	75
	Q ~ IGE ~ %	76	71	74	69	73	67	70	64
7500	GW ~ 100 LB	176	169	173	165	170	160	165	155
	Q ~ OGE ~ %	87	82	85	79	83	77	81	73
	Q ~ IGE ~ %	74	70	73	68	71	66	69	63
8000	GW ~ 100 LB	173	166	169	162	166	157	162	152
	Q ~ OGE ~ %	85	81	84	78	82	75	79	72
	Q ~ IGE ~ %	73	69	71	67	70	64	67	62
8500	GW ~ 100 LB	169	163	166	158	163	154	158	148
	Q ~ OGE ~ %	84	79	82	76	80	73	77	70
	Q ~ IGE ~ %	71	67	70	65	68	63	66	61
9000	GW ~ 100 LB	166	160	162	155	159	151	154	145
	Q ~ OGE ~ %	82	77	80	75	78	72	76	69
	Q ~ IGE ~ %	70	66	68	64	67	62	64	59
9500	GW ~ 100 LB	162	156	159	152	156	148	151	142
	Q ~ OGE ~ %	80	76	79	73	77	70	74	67
	Q ~ IGE ~ %	68	65	67	63	65	60	63	58
10,000	GW ~ 100 LB	159	153	155	149	152	145	148	139
	Q ~ OGE ~ %	79	74	77	72	75	69	72	66
	Q ~ IGE ~ %	67	63	66	61	64	59	62	57
10,500	GW ~ 100 LB	155	150	152	146	149	142	144	136
	Q ~ OGE ~ %	77	73	75	70	74	67	71	64
	Q ~ IGE ~ %	66	62	64	60	63	58	61	56

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9979_2_CL
SAF

Figure 4. Hover Torque Required (Sheet 11 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
11,000	GW ~ 100 LB	152	147	149	143	146	139	141	134
	Q ~ OGE ~ %	75	71	74	69	72	66	69	63
	Q ~ IGE ~ %	64	61	63	59	61	57	59	54
11,500	GW ~ 100 LB	149	144	146	140	143	136	138	131
	Q ~ OGE ~ %	74	70	72	67	70	65	68	62
	Q ~ IGE ~ %	63	60	62	58	60	56	58	53
12,000	GW ~ 100 LB	146	141	143	137	140	133	135	128
	Q ~ OGE ~ %	72	68	71	66	69	63	66	60
	Q ~ IGE ~ %	62	58	60	56	59	54	57	52
12,500	GW ~ 100 LB	143	138	140	134	137	130	133	125
	Q ~ OGE ~ %	71	67	69	64	67	62	65	59
	Q ~ IGE ~ %	60	57	59	55	58	53	55	51
13,000	GW ~ 100 LB	140	135	137	131	134	127	130	122
	Q ~ OGE ~ %	69	65	67	63	66	60	63	57
	Q ~ IGE ~ %	59	56	58	54	56	52	54	50
13,500	GW ~ 100 LB	137	132	135	128	131	124	127	119
	Q ~ OGE ~ %	68	64	66	61	64	59	62	56
	Q ~ IGE ~ %	58	55	56	53	55	51	53	48
14,000	GW ~ 100 LB	135	129	132	126	129	121	125	117
	Q ~ OGE ~ %	66	62	64	60	63	58	60	55
	Q ~ IGE ~ %	56	53	55	52	54	50	52	47
14,500	GW ~ 100 LB	132	127	129	123	126	119	122	114
	Q ~ OGE ~ %	65	61	63	59	61	56	59	53
	Q ~ IGE ~ %	55	52	54	50	52	48	50	46
15,000	GW ~ 100 LB	129	124	126	120	123	116	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	63	60	62	57	60	55		
	Q ~ IGE ~ %	54	51	53	49	51	47		
15,500	GW ~ 100 LB	126	121	123	117	120	113		
	Q ~ OGE ~ %	62	58	60	56	58	54		
	Q ~ IGE ~ %	53	50	51	48	50	46		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9979_3_CL
SAF

Figure 4. Hover Torque Required (Sheet 12 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		30		35		40		45	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	208	200	207	192	200	185	192	178
	Q ~ OGE ~ %	100	94	100	90	95	86	91	82
	Q ~ IGE ~ %	86	81	86	78	82	74	78	71
500	GW ~ 100 LB	207	196	204	189	196	182	188	175
	Q ~ OGE ~ %	100	93	98	88	94	84	89	80
	Q ~ IGE ~ %	86	80	84	76	81	73	77	70
1000	GW ~ 100 LB	206	193	200	186	193	179	185	171
	Q ~ OGE ~ %	100	91	97	87	92	83	87	79
	Q ~ IGE ~ %	86	78	83	75	79	72	75	68
1500	GW ~ 100 LB	203	189	196	182	189	176	182	168
	Q ~ OGE ~ %	99	90	95	85	91	82	86	77
	Q ~ IGE ~ %	85	77	81	74	78	71	74	67
2000	GW ~ 100 LB	200	186	193	179	186	172	178	165
	Q ~ OGE ~ %	97	88	93	84	89	80	84	76
	Q ~ IGE ~ %	83	76	80	72	76	69	73	66
2500	GW ~ 100 LB	196	183	189	176	182	169	175	162
	Q ~ OGE ~ %	96	86	91	82	87	79	83	75
	Q ~ IGE ~ %	82	74	78	71	75	68	71	65
3000	GW ~ 100 LB	193	179	186	173	179	166	172	159
	Q ~ OGE ~ %	94	85	90	81	86	77	81	73
	Q ~ IGE ~ %	80	73	77	70	74	67	70	63
3500	GW ~ 100 LB	189	176	182	169	175	163	168	156
	Q ~ OGE ~ %	92	83	88	79	84	75	80	72
	Q ~ IGE ~ %	79	72	76	68	72	65	69	62
4000	GW ~ 100 LB	185	173	179	166	172	160	165	153
	Q ~ OGE ~ %	90	82	86	78	82	74	78	70
	Q ~ IGE ~ %	77	70	74	67	71	64	67	61
4500	GW ~ 100 LB	182	169	175	163	169	156	162	150
	Q ~ OGE ~ %	89	80	85	76	81	72	76	69
	Q ~ IGE ~ %	76	69	73	66	69	62	66	60
5000	GW ~ 100 LB	178	166	172	159	165	153	158	146
	Q ~ OGE ~ %	87	78	83	74	79	71	75	67
	Q ~ IGE ~ %	74	67	71	64	68	61	64	58
5500	GW ~ 100 LB	174	162	168	156	162	150	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	85	77	81	73	77	69		
	Q ~ IGE ~ %	73	66	69	63	66	60		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9980_1_CL
SAF

Figure 4. Hover Torque Required (Sheet 13 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		30		35		40		45	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
6000	GW ~ 100 LB	171	159	164	153	158	147	151	140
	Q ~ OGE ~ %	83	75	79	71	75	68	71	64
	Q ~ IGE ~ %	71	64	68	61	65	59	61	56
6500	GW ~ 100 LB	167	156	161	150	155	144	148	137
	Q ~ OGE ~ %	81	73	77	70	74	66	70	63
	Q ~ IGE ~ %	70	63	66	60	63	57	60	54
7000	GW ~ 100 LB	164	153	158	146	152	141		
	Q ~ OGE ~ %	80	72	76	68	72	65		
	Q ~ IGE ~ %	68	62	65	59	62	56		
7500	GW ~ 100 LB	161	149	155	143	148	138		
	Q ~ OGE ~ %	78	70	74	67	70	63		
	Q ~ IGE ~ %	67	61	64	58	61	55		
8000	GW ~ 100 LB	157	146	151	140	146	135		
	Q ~ OGE ~ %	76	69	73	65	69	62		
	Q ~ IGE ~ %	65	59	62	56	59	54		
8500	GW ~ 100 LB	154	143	148	137	142	132		
	Q ~ OGE ~ %	75	67	71	64	67	61		
	Q ~ IGE ~ %	64	58	61	55	58	53		
9000	GW ~ 100 LB	151	140	145	134				
	Q ~ OGE ~ %	73	66	69	62	ABOVE ENGINE AMBIENT TEMP LIMIT			
	Q ~ IGE ~ %	63	57	60	54				
9500	GW ~ 100 LB	148	137	142	132				
	Q ~ OGE ~ %	71	64	68	61				
	Q ~ IGE ~ %	61	56	58	53				
10,000	GW ~ 100 LB	144	134	139	129				
	Q ~ OGE ~ %	70	63	66	60				
	Q ~ IGE ~ %	60	54	57	52				
10,500	GW ~ 100 LB	141	131	136	126				
	Q ~ OGE ~ %	68	62	65	58				
	Q ~ IGE ~ %	58	53	56	50				
11,000	GW ~ 100 LB	138	129						
	Q ~ OGE ~ %	67	60						
	Q ~ IGE ~ %	57	52						
11,500	GW ~ 100 LB	135	126						
	Q ~ OGE ~ %	65	59						
	Q ~ IGE ~ %	56	51						

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9980_2_CL
SAF

Figure 4. Hover Torque Required (Sheet 14 of 15) 700

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-700 ENGINE (CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C			
		50		55	
		1.0	0.9	1.0	0.9
0	GW ~ 100 LB	184	170	176	163
	Q ~ OGE ~ %	86	77	82	73
	Q ~ IGE ~ %	74	67	71	64
500	GW ~ 100 LB	180	167	173	160
	Q ~ OGE ~ %	84	76	80	72
	Q ~ IGE ~ %	73	66	69	63
1000	GW ~ 100 LB	177	164	170	157
	Q ~ OGE ~ %	83	75	79	71
	Q ~ IGE ~ %	72	65	68	62
1500	GW ~ 100 LB	174	161	167	154
	Q ~ OGE ~ %	81	73	77	69
	Q ~ IGE ~ %	70	64	67	61
2000	GW ~ 100 LB	171	158	163	151
	Q ~ OGE ~ %	80	72	76	68
	Q ~ IGE ~ %	69	62	65	59
2500	GW ~ 100 LB	167	155	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	78	70		
	Q ~ IGE ~ %	68	61		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

100% Q TRANSMISSION LIMIT OR

30 MINUTE ENGINE Q LIMIT.

AA9981CL
SAF

Figure 4. Hover Torque Required (Sheet 15 of 15) 700

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% -% RPM R = 100%) PRESSURE ALTITUDE - FEET					
	-1000	-500	0	500	1000	1500
55	781	786	789	793	797	802
50	768	772	776	780	784	788
45	754	759	762	766	770	774
41	744	748	751	755	759	763
39	738	742	746	750	754	758
37	733	737	740	744	748	752
35	728	732	735	739	743	747
33	722	726	730	733	737	741
31	717	721	724	728	732	735
29	711	716	719	723	726	730
27	706	710	713	717	721	725
25	701	705	708	712	715	719
23	695	699	703	706	710	714
21	690	694	697	701	704	708
19	684	688	692	695	699	703
17	679	683	686	690	693	697
15	673	677	680	684	688	692
13	668	672	675	679	682	686
11	663	667	670	673	676	681
9	657	661	664	668	671	675
7	652	656	659	662	666	670
5	647	651	654	657	660	665
3	642	645	648	652	655	659
1	636	640	643	647	650	654
-1	631	635	638	641	645	649
-3	626	630	633	636	640	643
-5	621	625	628	631	634	638
-7	616	619	623	626	629	633
-9	610	614	617	621	624	628
-11	605	609	612	615	619	623
-13	600	604	607	610	613	617
-15	594	598	601	605	608	612
-17	589	593	596	599	603	606
-19	584	587	590	594	597	601
-21	578	582	585	588	592	595
-25	568	571	574	578	581	585
-30	554	558	561	564	567	571
-35	541	545	548	551	554	558
-40	528	532	534	538	541	544
-45	515	519	522	525	528	531
-50	502	506	508	511	514	518
-55	489	493	495	498	501	504

AB2698_1CL
SA

Figure 5. TGT Reference Table (Sheet 1 of 3)

700

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% -% RPM R = 100%) PRESSURE ALTITUDE - FEET					
	2000	2500	3000	3500	4000	4500
55	806	811	815	821	826	832
50	792	797	801	807	812	817
45	778	783	787	793	797	803
41	767	772	776	781	785	791
39	762	766	770	776	780	786
37	756	760	765	770	775	780
35	750	755	759	764	769	774
33	745	749	753	759	763	768
31	739	744	748	753	758	763
29	734	738	742	748	752	757
27	728	732	737	742	746	751
25	723	727	731	736	741	746
23	717	721	726	731	735	740
21	712	716	720	725	729	734
19	706	710	714	720	724	728
17	701	705	709	714	718	723
15	695	699	703	708	712	717
13	690	694	698	703	707	711
11	684	688	692	697	701	705
9	679	683	687	691	695	700
7	673	677	681	686	689	694
5	668	672	676	680	684	688
3	663	667	670	675	678	683
1	657	661	665	669	673	677
-1	652	656	660	664	667	672
-3	647	651	654	659	662	666
-5	642	645	649	653	657	661
-7	636	640	644	648	651	656
-9	631	635	639	643	646	650
-11	626	630	633	637	641	645
-13	621	624	628	632	636	640
-15	615	619	622	627	630	634
-17	610	614	617	621	625	629
-19	604	608	611	616	619	623
-21	599	603	606	610	614	618
-25	588	591	595	599	602	606
-30	574	578	581	585	588	592
-35	560	564	567	571	575	578
-40	547	550	554	557	561	564
-45	534	537	540	544	547	551
-50	520	524	527	531	534	537
-55	507	510	513	517	520	524

AB2698_2CL
SA

Figure 5. TGT Reference Table (Sheet 2 of 3) **700**

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% -% RPM R = 100%) PRESSURE ALTITUDE - FEET					
	5000	6000	7000	8000	9000	10000
55						
50	822	833				
45	808	819	830			
41	796	807	818	830		
39	791	801	812	824		
37	785	795	806	818	830	
35	779	790	800	812	824	
33	773	784	794	806	818	831
31	767	778	788	800	812	825
29	762	772	782	794	805	818
27	756	766	777	788	799	812
25	750	760	771	782	793	806
23	744	754	765	776	787	800
21	739	745	759	770	781	793
19	733	743	753	764	775	787
17	727	737	747	758	769	781
15	721	731	741	752	763	775
13	716	725	735	746	757	768
11	710	719	729	740	751	762
9	704	713	723	734	745	756
7	698	708	717	728	738	750
5	692	702	711	722	732	744
3	687	696	705	716	726	737
1	681	690	699	709	720	731
-1	676	684	693	703	714	725
-3	670	679	687	697	707	718
-5	665	673	682	692	701	712
-7	659	668	676	686	696	706
-9	654	662	670	680	690	700
-11	649	657	665	674	684	694
-13	643	651	659	668	678	688
-15	638	646	654	663	672	682
-17	632	640	648	657	666	676
-19	627	635	643	652	661	670
-21	621	629	637	646	655	665
-25	610	618	626	635	644	653
-30	596	604	612	620	629	638
-35	582	590	597	606	614	623
-40	568	575	583	592	600	609
-45	554	561	569	578	586	594
-50	540	546	555	563	571	579
-55	527	534	541	549	557	565

AB2698_3CL
SA

Figure 5. TGT Reference Table (Sheet 3 of 3)

700

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-50	-45	-40	-35	-30	-25	-20
0	1.0	129	130	131	132	133	134	135
	0.9	125	126	127	128	129	130	131
500	1.0	127	128	129	130	131	131	132
	0.9	123	124	125	126	127	128	128
1000	1.0	124	125	126	127	128	129	130
	0.9	121	122	123	124	125	125	126
1500	1.0	122	123	124	125	126	127	128
	0.9	119	120	120	121	122	123	124
2000	1.0	120	121	122	123	124	125	125
	0.9	116	117	118	119	120	121	122
2500	1.0	118	119	120	121	122	122	123
	0.9	114	115	116	117	118	119	120
3000	1.0	116	117	118	119	120	120	121
	0.9	112	113	114	115	116	117	117
3500	1.0	114	114	115	116	117	118	119
	0.9	110	111	112	113	114	115	115
4000	1.0	112	112	113	114	115	116	117
	0.9	109	109	110	111	112	113	114
4500	1.0	110	110	111	112	113	114	115
	0.9	107	107	108	109	110	111	112
5000	1.0	108	108	109	110	111	112	112
	0.9	105	105	106	107	108	109	109
5500	1.0	106	106	107	108	109	110	110
	0.9	103	103	104	105	106	107	107
6000	1.0	104	104	105	106	107	108	108
	0.9	101	101	102	103	104	105	105
6500	1.0	102	103	103	104	105	106	106
	0.9	99	100	100	101	102	103	103
7000	1.0	100	101	101	102	103	104	104
	0.9	97	98	98	99	100	101	101
7500	1.0	98	99	100	100	101	102	102
	0.9	95	96	97	97	98	99	99

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

AB3668_1_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 1 of
6) 701C 701D/CC**

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-50	-45	-40	-35	-30	-25	-20
8000	1.0	96	97	98	99	99	100	101
	0.9	93	94	95	96	96	97	98
8500	1.0	94	95	96	97	98	98	99
	0.9	91	92	93	94	95	95	96
9000	1.0	93	93	94	95	96	96	97
	0.9	90	90	91	92	93	93	94
9500	1.0	91	92	92	93	94	94	95
	0.9	88	89	89	90	91	91	92
10,000	1.0	89	90	91	91	92	93	93
	0.9	86	87	88	88	89	90	90
10,500	1.0	88	88	89	90	90	91	91
	0.9	85	85	86	87	87	88	88
11,000	1.0	86	87	87	88	89	89	90
	0.9	84	84	84	85	86	86	87
11,500	1.0	84	85	86	86	87	87	88
	0.9	82	83	84	84	84	84	85
12,000	1.0	83	83	84	85	85	86	86
	0.9	81	81	82	83	83	84	84
12,500	1.0	81	82	82	83	83	84	85
	0.9	79	80	80	81	81	82	83
13,000	1.0	80	80	81	81	82	82	83
	0.9	78	78	79	79	80	80	81
13,500	1.0	78	78	79	80	80	81	81
	0.9	76	76	77	78	78	79	79
14,000	1.0	76	77	77	78	79	79	80
	0.9	74	75	75	76	77	77	78
14,500	1.0	75	75	76	76	77	77	78
	0.9	73	73	74	74	75	75	76
15,000	1.0	73	74	74	75	75	76	76
	0.9	71	72	72	73	73	74	74
15,500	1.0	72	72	73	73	74	74	75
	0.9	70	70	71	71	72	72	73

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

AB3668_2_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 2 of
6) 701C 701D/CC**

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-15	-10	-5	0	5	10	15
0	1.0	136	138	137	136	134	132	131
	0.9	132	134	133	132	129	126	123
500	1.0	134	135	135	134	132	130	128
	0.9	130	131	131	129	127	124	121
1000	1.0	131	133	132	131	130	128	126
	0.9	128	129	128	127	124	122	119
1500	1.0	129	131	130	129	127	125	124
	0.9	125	127	126	125	122	119	117
2000	1.0	127	128	127	127	125	123	121
	0.9	123	124	124	123	120	117	115
2500	1.0	124	126	125	124	123	121	119
	0.9	121	122	121	120	118	115	113
3000	1.0	122	124	123	122	120	119	117
	0.9	119	120	119	118	116	113	110
3500	1.0	120	122	121	120	118	116	115
	0.9	117	118	117	116	113	111	108
4000	1.0	118	119	119	118	116	114	113
	0.9	115	116	116	114	111	109	107
4500	1.0	116	117	116	115	114	112	111
	0.9	113	114	113	111	109	107	105
5000	1.0	114	115	114	113	112	110	109
	0.9	111	112	111	109	108	105	103
5500	1.0	112	113	112	111	110	108	106
	0.9	109	110	109	107	106	103	100
6000	1.0	110	111	110	109	108	106	104
	0.9	107	108	107	105	104	101	98
6500	1.0	108	109	108	107	105	104	102
	0.9	105	106	105	103	101	99	96
7000	1.0	106	107	106	105	103	102	100
	0.9	103	104	103	102	99	97	94
7500	1.0	104	105	104	103	101	100	98
	0.9	101	102	101	100	97	95	93

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

AB3668_3_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 3 of
6) 701C 701D/CC**

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		-15	-10	-5	0	5	10	15
8000	1.0	102	103	102	101	99	98	96
	0.9	99	100	99	98	95	93	91
8500	1.0	100	101	100	99	97	96	94
	0.9	97	98	97	96	93	91	89
9000	1.0	98	99	98	97	95	94	92
	0.9	95	96	95	94	91	89	87
9500	1.0	96	97	96	95	93	92	91
	0.9	93	94	93	92	89	88	86
10,000	1.0	94	95	94	93	92	90	89
	0.9	91	92	91	90	88	86	84
10,500	1.0	92	94	92	91	90	88	87
	0.9	89	91	89	88	86	84	82
11,000	1.0	91	92	91	89	88	87	85
	0.9	88	89	88	86	84	83	80
11,500	1.0	89	90	89	88	86	85	83
	0.9	86	87	86	85	83	81	78
12,000	1.0	87	88	87	86	84	83	82
	0.9	84	85	84	83	81	79	77
12,500	1.0	86	87	85	84	83	81	80
	0.9	84	84	83	81	80	77	76
13,000	1.0	84	85	84	82	81	80	78
	0.9	82	83	82	79	78	76	74
13,500	1.0	82	83	82	81	79	78	77
	0.9	80	81	80	78	76	74	73
14,000	1.0	81	82	80	79	78	76	75
	0.9	79	80	78	76	75	72	71
14,500	1.0	79	80	79	77	76	75	73
	0.9	77	78	77	74	73	71	69
15,000	1.0	77	78	77	76	75	73	72
	0.9	75	76	75	73	72	69	68
15,500	1.0	76	76	75	74	73	72	71
	0.9	74	74	73	72	70	69	67

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

AB3668_4_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 4 of
6) 701C 701D/CC**

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		20	25	30	35	40	45	50
0	1.0	129	127	125	121	117	113	109
	0.9	120	117	113	109	106	102	98
500	1.0	127	125	123	119	115	111	107
	0.9	118	115	111	107	104	100	96
1000	1.0	124	122	120	117	113	109	105
	0.9	116	113	109	105	102	98	94
1500	1.0	122	120	118	114	111	107	103
	0.9	114	111	107	103	100	96	93
2000	1.0	120	118	116	112	109	105	101
	0.9	112	109	105	101	98	94	91
2500	1.0	118	116	114	110	107	103	99
	0.9	110	107	104	99	96	93	89
3000	1.0	116	114	112	108	105	101	97
	0.9	108	105	102	97	95	91	87
3500	1.0	113	111	109	106	102	99	95
	0.9	105	102	99	95	92	89	86
4000	1.0	111	109	107	104	100	97	93
	0.9	104	100	97	94	90	87	84
4500	1.0	109	107	105	102	98	95	91
	0.9	102	99	95	92	88	86	82
5000	1.0	107	105	103	100	96	93	89
	0.9	100	97	94	90	86	84	80
5500	1.0	105	103	101	98	94	91	87
	0.9	98	95	92	88	85	82	78
6000	1.0	103	101	99	96	92	89	85
	0.9	96	93	90	86	83	80	77
6500	1.0	101	99	97	94	90	87	83
	0.9	94	91	88	85	81	78	75
7000	1.0	99	97	95	92	89	85	82
	0.9	92	89	86	83	80	77	74
7500	1.0	97	95	93	90	87	83	80
	0.9	91	88	85	81	78	75	72

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

AREA IS ABOVE ENGINE AMBIENT TEMPERATURE LIMIT

AB3668_5_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 5 of
6) 701C 701D/CC**

**MAXIMUM TORQUE AVAILABLE* – 10 MINUTE LIMIT
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**



HP ~FT	ATF	FREE AIR TEMPERATURE °C						
		20	25	30	35	40	45	50
8000	1.0	95	93	91	88	85	81	78
	0.9	89	86	83	79	77	73	70
8500	1.0	93	91	89	86	83	80	76
	0.9	87	84	81	77	75	72	68
9000	1.0	91	89	87	84	81	78	74
	0.9	85	82	79	76	73	70	67
9500	1.0	89	87	85	82	79	76	73
	0.9	83	80	77	74	71	68	66
10,000	1.0	87	85	83	80	78	74	71
	0.9	81	78	75	72	70	67	64
10,500	1.0	85	84	82	79	76	73	69
	0.9	79	77	75	71	68	66	62
11,000	1.0	84	82	80	77	74	71	68
	0.9	78	76	73	69	67	64	61
11,500	1.0	82	80	78	75	72	69	66
	0.9	77	74	71	68	65	62	59
12,000	1.0	80	78	76	74	71	68	64
	0.9	75	72	69	67	64	61	58
12,500	1.0	78	77	75	72	69	66	63
	0.9	73	71	68	65	62	59	57
13,000	1.0	77	75	73	70	68	65	61
	0.9	72	69	66	63	61	59	55
13,500	1.0	75	73	71	69	66	63	60
	0.9	70	67	65	62	59	57	54
14,000	1.0	74	72	70	67	65	62	59
	0.9	69	66	64	60	59	56	53
14,500	1.0	72	70	68	66	63	60	57
	0.9	67	65	62	59	57	54	51
15,000	1.0	71	69	67	64	62	59	56
	0.9	66	64	61	58	56	53	50
15,500	1.0	69	67	65	63	60	57	54
	0.9	64	62	59	57	54	51	49

* MAXIMUM TORQUE AVAILABLE IS LIMITED TO 120%
BELOW 80 KIAS AND 100% TORQUE ABOVE 80 KIAS

■ AREA IS ABOVE ENGINE AMBIENT TEMPERATURE LIMIT

AB3668_6_CL
SAF

**Figure 6. Maximum Torque Available (Sheet 6 of
6) 701C 701D/CC**

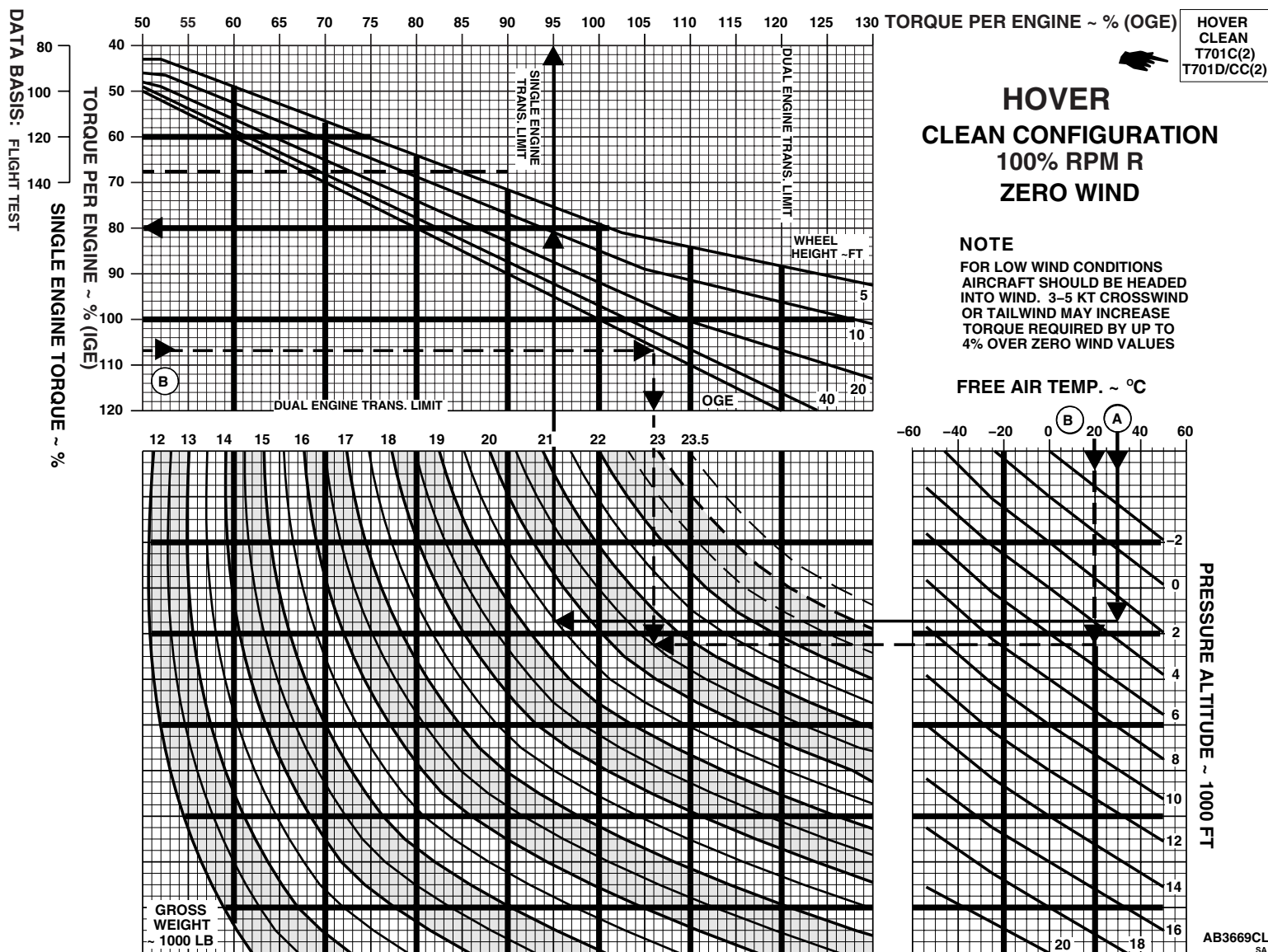


Figure 7. Hover - Clean Configuration

701C **701D/CC**

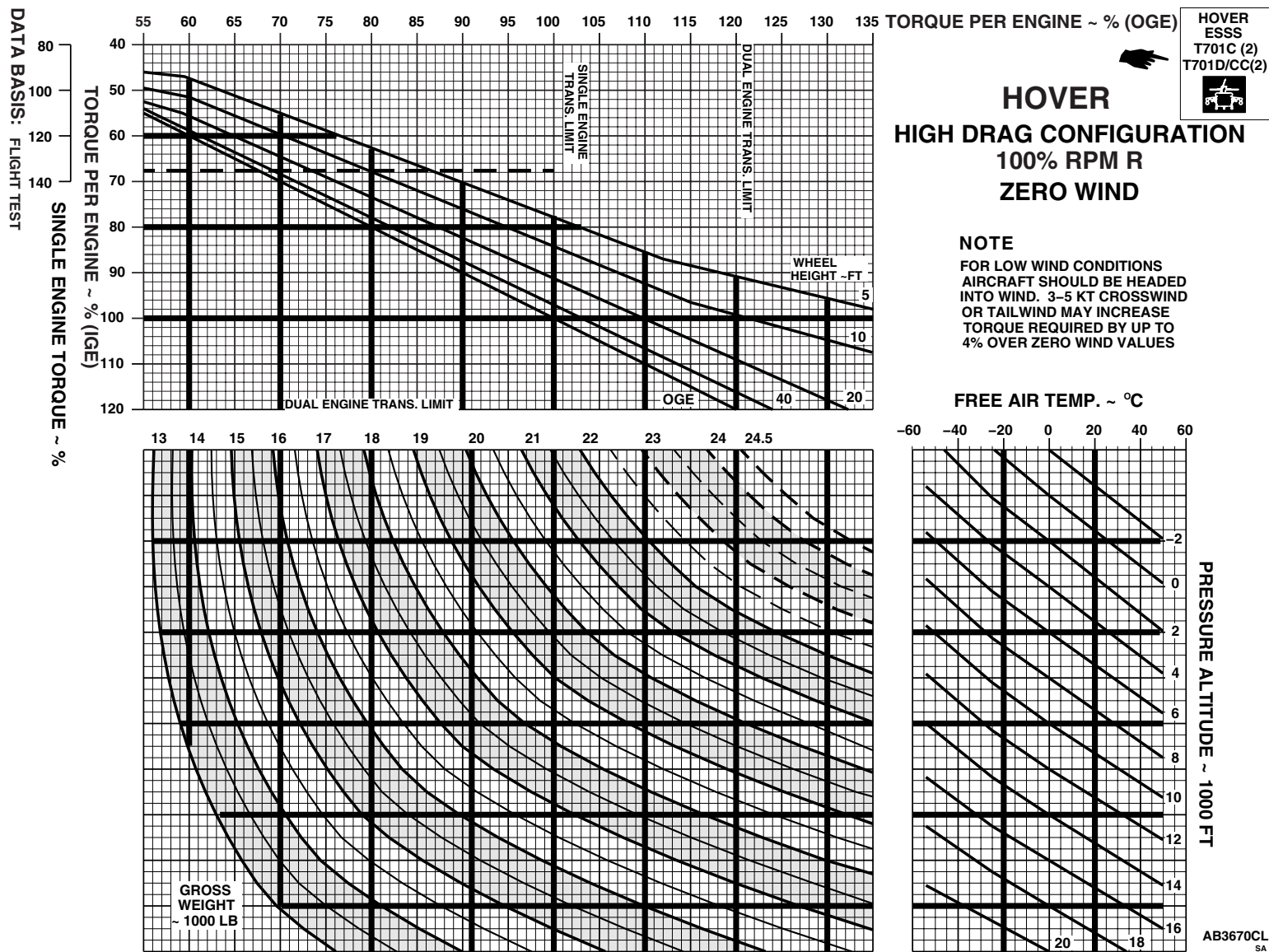


Figure 8. Hover - High Drag Configuration **701C** **701D/CC**

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	97	97	98	98	99	99
	Q ~ IGE ~ %	84	84	85	85	85	85
1000	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	99	99	99	99	100	100
	Q ~ IGE ~ %	85	85	85	85	86	86
2000	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	100	100	101	101	102	102
	Q ~ IGE ~ %	86	86	86	86	86	86
3000	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	102	102	103	103	104	104
	Q ~ IGE ~ %	86	86	87	87	87	87
4000	GW ~ 100 LB	220	220	220	220	220	220
	Q ~ OGE ~ %	104	104	105	105	106	106
	Q ~ IGE ~ %	87	87	88	88	88	88
5000	GW ~ 100 LB	220	218	220	217	220	217
	Q ~ OGE ~ %	107	105	108	106	109	107
	Q ~ IGE ~ %	88	87	89	87	89	88
5500	GW ~ 100 LB	217	214	217	213	217	213
	Q ~ OGE ~ %	106	103	107	104	108	105
	Q ~ IGE ~ %	87	85	88	86	88	86
6000	GW ~ 100 LB	213	210	213	209	213	209
	Q ~ OGE ~ %	104	101	105	102	106	103
	Q ~ IGE ~ %	86	84	86	84	87	85
6500	GW ~ 100 LB	209	206	209	206	209	206
	Q ~ OGE ~ %	102	100	103	100	104	101
	Q ~ IGE ~ %	84	82	85	83	85	83

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3671_1_CL
SAF

Figure 9. Hover Torque Required (Sheet 1 of 15)

701C **701D/CC**

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
7000	GW ~ 100 LB	205	202	205	202	205	202
	Q ~ OGE ~ %	101	98	101	98	102	99
	Q ~ IGE ~ %	83	81	83	81	84	82
7500	GW ~ 100 LB	202	198	201	198	201	198
	Q ~ OGE ~ %	99	96	99	97	100	97
	Q ~ IGE ~ %	81	79	82	80	82	80
8000	GW ~ 100 LB	198	195	198	194	198	194
	Q ~ OGE ~ %	97	94	98	95	98	96
	Q ~ IGE ~ %	79	78	80	78	81	79
8500	GW ~ 100 LB	194	191	194	191	194	191
	Q ~ OGE ~ %	95	92	96	93	97	94
	Q ~ IGE ~ %	78	76	79	77	79	77
9000	GW ~ 100 LB	191	187	190	187	190	187
	Q ~ OGE ~ %	93	91	94	91	95	92
	Q ~ IGE ~ %	77	75	77	75	78	76
9500	GW ~ 100 LB	187	184	187	184	187	184
	Q ~ OGE ~ %	92	89	92	90	93	90
	Q ~ IGE ~ %	75	73	76	74	76	74
10,000	GW ~ 100 LB	183	180	183	180	183	180
	Q ~ OGE ~ %	90	87	91	88	91	89
	Q ~ IGE ~ %	74	72	74	72	75	73
10,500	GW ~ 100 LB	180	177	180	177	180	177
	Q ~ OGE ~ %	88	86	89	86	90	87
	Q ~ IGE ~ %	72	71	73	71	73	72
11,000	GW ~ 100 LB	176	174	176	173	176	173
	Q ~ OGE ~ %	87	84	87	85	88	85
	Q ~ IGE ~ %	71	69	71	70	72	70

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3671_2_CL
SAF

Figure 9. Hover Torque Required (Sheet 2 of 15)

701C

701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C					
		-45		-40		-35	
		1.0	0.9	1.0	0.9	1.0	0.9
11,500	GW ~ 100 LB	173	170	173	170	173	170
	Q ~ OGE ~ %	85	82	86	83	86	84
	Q ~ IGE ~ %	70	68	70	68	70	69
12,000	GW ~ 100 LB	170	167	170	167	169	167
	Q ~ OGE ~ %	83	81	84	81	84	82
	Q ~ IGE ~ %	68	67	69	67	69	68
12,500	GW ~ 100 LB	166	164	166	164	166	163
	Q ~ OGE ~ %	82	79	82	80	83	80
	Q ~ IGE ~ %	67	65	67	66	68	66
13,000	GW ~ 100 LB	163	160	163	160	163	160
	Q ~ OGE ~ %	80	78	81	78	81	79
	Q ~ IGE ~ %	66	64	66	65	66	65
13,500	GW ~ 100 LB	160	157	160	157	160	157
	Q ~ OGE ~ %	78	76	79	77	79	77
	Q ~ IGE ~ %	64	63	65	63	65	64
14,000	GW ~ 100 LB	157	154	157	154	156	154
	Q ~ OGE ~ %	77	75	77	75	78	76
	Q ~ IGE ~ %	63	62	63	62	64	62
14,500	GW ~ 100 LB	154	151	153	151	153	151
	Q ~ OGE ~ %	75	73	76	74	76	74
	Q ~ IGE ~ %	62	60	62	61	62	61
15,000	GW ~ 100 LB	151	148	150	148	150	148
	Q ~ OGE ~ %	74	72	74	72	75	73
	Q ~ IGE ~ %	61	59	61	60	61	60
15,500	GW ~ 100 LB	148	145	147	145	147	145
	Q ~ OGE ~ %	72	70	73	71	73	71
	Q ~ IGE ~ %	59	58	60	58	60	59

* GW ~ 100 LB = GW DIVIDED BY 100 LB
Q = TORQUE %
OGE ≅ 100 FT WHEEL HEIGHT
IGE = 10 FT WHEEL HEIGHT
MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB
120% Q TRANSMISSION LIMIT OR
10 MINUTE ENGINE Q LIMIT.

AB3671_3_CL
SAF

Figure 9. Hover Torque Required (Sheet 3 of 15)

701C **701D/CC**

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	99	99	100	100	101	101	101	101
	Q ~ IGE ~ %	85	85	86	86	86	86	87	87
1000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	101	101	101	101	102	102	103	103
	Q ~ IGE ~ %	86	86	87	87	87	87	88	88
2000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	103	103	103	103	104	104	105	105
	Q ~ IGE ~ %	87	87	87	87	88	88	89	89
3000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	105	105	106	106	106	106	107	107
	Q ~ IGE ~ %	88	88	88	88	89	89	90	90
4000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	107	107	108	108	109	109	109	109
	Q ~ IGE ~ %	89	89	90	90	90	90	91	91
5000	GW ~ 100 LB	220	218	220	217	220	217	220	218
	Q ~ OGE ~ %	110	108	111	108	111	109	112	110
	Q ~ IGE ~ %	90	89	91	89	92	90	92	91
5500	GW ~ 100 LB	217	214	217	213	217	213	217	214
	Q ~ OGE ~ %	109	106	109	106	110	107	111	108
	Q ~ IGE ~ %	89	87	90	88	91	88	92	89
6000	GW ~ 100 LB	213	210	213	209	213	209	213	210
	Q ~ OGE ~ %	107	104	107	104	108	105	109	106
	Q ~ IGE ~ %	88	86	88	86	89	87	90	88
6500	GW ~ 100 LB	209	206	209	205	209	205	209	206
	Q ~ OGE ~ %	105	102	105	102	106	103	107	104
	Q ~ IGE ~ %	86	84	86	84	87	85	88	86

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3672_1_CL
SAF

Figure 9. Hover Torque Required (Sheet 4 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
7000	GW ~ 100 LB	206	202	205	201	205	202	206	202
	Q ~ OGE ~ %	103	100	103	100	104	101	105	102
	Q ~ IGE ~ %	84	82	85	83	86	84	87	85
7500	GW ~ 100 LB	202	198	201	198	201	198	202	198
	Q ~ OGE ~ %	101	98	101	99	102	99	103	101
	Q ~ IGE ~ %	83	81	83	81	84	82	85	83
8000	GW ~ 100 LB	198	194	197	194	198	194	198	195
	Q ~ OGE ~ %	99	96	100	97	100	98	102	99
	Q ~ IGE ~ %	81	79	82	80	82	80	83	81
8500	GW ~ 100 LB	194	191	194	190	194	191	194	191
	Q ~ OGE ~ %	98	95	98	95	99	96	100	97
	Q ~ IGE ~ %	80	78	80	78	81	79	82	80
9000	GW ~ 100 LB	191	187	190	187	190	187	191	187
	Q ~ OGE ~ %	96	93	96	93	97	94	98	95
	Q ~ IGE ~ %	78	76	79	77	79	78	80	78
9500	GW ~ 100 LB	187	184	187	183	187	184	187	184
	Q ~ OGE ~ %	94	91	94	91	95	92	96	93
	Q ~ IGE ~ %	77	75	77	75	78	76	79	77
10,000	GW ~ 100 LB	183	180	183	180	183	180	183	180
	Q ~ OGE ~ %	92	89	92	90	93	90	94	91
	Q ~ IGE ~ %	75	74	76	74	76	75	77	75
10,500	GW ~ 100 LB	180	177	180	176	180	177	180	177
	Q ~ OGE ~ %	90	88	91	88	91	89	92	90
	Q ~ IGE ~ %	74	72	74	73	75	73	76	74
11,000	GW ~ 100 LB	176	173	176	173	176	173	176	173
	Q ~ OGE ~ %	89	86	89	86	90	87	90	88
	Q ~ IGE ~ %	73	71	73	71	74	72	74	73

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3672_2_CL
SAF

Figure 9. Hover Torque Required (Sheet 5 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-30		-25		-20		-15	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
11,500	GW ~ 100 LB	173	170	173	170	173	170	173	170
	Q ~ OGE ~ %	87	84	87	85	88	85	89	86
	Q ~ IGE ~ %	71	69	71	70	72	70	73	71
12,000	GW ~ 100 LB	170	167	169	166	169	166	170	167
	Q ~ OGE ~ %	85	83	85	83	86	84	87	84
	Q ~ IGE ~ %	70	68	70	68	71	69	71	70
12,500	GW ~ 100 LB	166	163	166	163	166	163	166	164
	Q ~ OGE ~ %	83	81	84	81	85	82	85	83
	Q ~ IGE ~ %	68	67	69	67	69	68	70	68
13,000	GW ~ 100 LB	163	160	163	160	163	160	163	161
	Q ~ OGE ~ %	82	79	82	80	83	80	84	81
	Q ~ IGE ~ %	67	65	67	66	68	66	69	67
13,500	GW ~ 100 LB	160	157	159	157	160	157	160	157
	Q ~ OGE ~ %	80	78	80	78	81	79	82	80
	Q ~ IGE ~ %	66	64	66	64	67	65	68	66
14,000	GW ~ 100 LB	156	154	156	154	156	154	157	154
	Q ~ OGE ~ %	78	76	79	77	80	77	81	78
	Q ~ IGE ~ %	64	63	65	63	65	64	66	65
14,500	GW ~ 100 LB	153	151	153	151	153	151	154	151
	Q ~ OGE ~ %	77	75	77	75	78	76	79	77
	Q ~ IGE ~ %	63	62	63	62	64	62	65	63
15,000	GW ~ 100 LB	150	148	150	148	150	148	151	148
	Q ~ OGE ~ %	75	73	76	74	76	74	78	75
	Q ~ IGE ~ %	62	60	62	61	63	61	64	62
15,500	GW ~ 100 LB	147	145	147	145	147	145	148	146
	Q ~ OGE ~ %	74	72	74	72	75	73	76	74
	Q ~ IGE ~ %	60	59	61	59	61	60	62	61

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3672_3_CL
SAF

Figure 9. Hover Torque Required (Sheet 6 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	102	102	102	102	103	103	104	104
	Q ~ IGE ~ %	87	87	88	88	88	88	89	89
2000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	106	106	106	106	107	107	108	108
	Q ~ IGE ~ %	89	89	90	90	90	90	91	91
3000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	108	108	108	108	109	109	110	110
	Q ~ IGE ~ %	90	90	91	91	92	92	92	92
4000	GW ~ 100 LB	220	220	220	220	220	220	220	218
	Q ~ OGE ~ %	110	110	111	111	112	112	113	111
	Q ~ IGE ~ %	92	92	92	92	93	93	94	93
5000	GW ~ 100 LB	220	218	220	217	218	214	215	210
	Q ~ OGE ~ %	113	112	114	112	113	110	112	107
	Q ~ IGE ~ %	93	92	94	92	94	91	93	90
5500	GW ~ 100 LB	218	214	217	213	214	210	211	206
	Q ~ OGE ~ %	113	110	113	110	111	108	110	105
	Q ~ IGE ~ %	93	91	93	91	92	89	91	88
6000	GW ~ 100 LB	214	210	213	209	210	206	207	203
	Q ~ OGE ~ %	111	108	111	108	109	106	108	103
	Q ~ IGE ~ %	91	89	91	89	90	88	89	86
6500	GW ~ 100 LB	210	207	209	205	206	202	204	199
	Q ~ OGE ~ %	109	106	109	106	107	104	106	101
	Q ~ IGE ~ %	89	87	89	87	88	86	88	85
7000	GW ~ 100 LB	206	203	205	202	202	198	200	195
	Q ~ OGE ~ %	107	104	107	104	105	102	104	99
	Q ~ IGE ~ %	88	86	88	86	87	84	86	83

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3673_1_CL
SAF

Figure 9. Hover Torque Required (Sheet 7 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
7500	GW ~ 100 LB	202	199	201	198	198	195	196	191
	Q ~ OGE ~ %	105	102	105	102	103	100	102	97
	Q ~ IGE ~ %	86	84	86	84	85	83	84	81
8000	GW ~ 100 LB	199	195	197	194	194	191	192	187
	Q ~ OGE ~ %	103	100	102	100	101	98	99	95
	Q ~ IGE ~ %	85	83	84	82	83	81	82	80
8500	GW ~ 100 LB	195	192	193	190	191	187	188	184
	Q ~ OGE ~ %	101	98	100	98	99	96	98	94
	Q ~ IGE ~ %	83	81	83	81	82	80	81	78
9000	GW ~ 100 LB	191	188	190	186	187	183	184	180
	Q ~ OGE ~ %	99	96	98	96	97	94	96	92
	Q ~ IGE ~ %	81	79	81	79	80	78	79	77
9500	GW ~ 100 LB	187	184	186	183	183	180	181	177
	Q ~ OGE ~ %	97	94	96	94	95	92	94	90
	Q ~ IGE ~ %	80	78	79	78	79	76	78	75
10,000	GW ~ 100 LB	184	181	182	179	180	176	177	173
	Q ~ OGE ~ %	95	92	94	92	93	90	92	88
	Q ~ IGE ~ %	78	76	78	76	77	75	76	73
10,500	GW ~ 100 LB	180	177	179	176	176	173	174	170
	Q ~ OGE ~ %	93	91	93	90	91	88	90	86
	Q ~ IGE ~ %	77	75	76	75	75	73	75	72
11,000	GW ~ 100 LB	177	174	175	172	173	169	170	166
	Q ~ OGE ~ %	92	89	91	88	89	86	88	85
	Q ~ IGE ~ %	75	74	75	73	74	72	73	71
11,500	GW ~ 100 LB	174	171	172	169	169	166	167	163
	Q ~ OGE ~ %	90	87	89	86	88	85	86	83
	Q ~ IGE ~ %	74	72	73	72	73	70	72	69

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3673_2_CL
SAF

Figure 9. Hover Torque Required (Sheet 8 of 15)

701C

701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		-10		-5		0		5	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
12,000	GW ~ 100 LB	170	167	168	165	166	163	164	160
	Q ~ OGE ~ %	88	86	87	85	86	83	85	81
	Q ~ IGE ~ %	72	71	72	70	71	69	70	68
12,500	GW ~ 100 LB	167	164	165	162	163	159	160	157
	Q ~ OGE ~ %	87	84	85	83	84	81	83	79
	Q ~ IGE ~ %	71	69	70	69	70	68	69	66
13,000	GW ~ 100 LB	164	161	162	159	159	156	157	153
	Q ~ OGE ~ %	85	83	84	81	82	80	81	78
	Q ~ IGE ~ %	70	68	69	67	68	66	67	65
13,500	GW ~ 100 LB	160	158	158	156	156	153	154	150
	Q ~ OGE ~ %	83	81	82	80	81	78	79	76
	Q ~ IGE ~ %	68	67	68	66	67	65	66	64
14,000	GW ~ 100 LB	157	155	155	153	153	150	151	147
	Q ~ OGE ~ %	82	79	80	78	79	76	78	75
	Q ~ IGE ~ %	67	65	66	65	65	64	65	62
14,500	GW ~ 100 LB	154	152	152	149	150	147	148	144
	Q ~ OGE ~ %	80	78	79	76	77	75	76	73
	Q ~ IGE ~ %	66	64	65	63	64	62	63	61
15,000	GW ~ 100 LB	151	148	149	146	147	144	145	141
	Q ~ OGE ~ %	78	76	77	75	76	73	75	72
	Q ~ IGE ~ %	64	63	64	62	63	61	62	60
15,500	GW ~ 100 LB	148	145	146	143	144	141	142	138
	Q ~ OGE ~ %	77	74	75	73	74	72	73	70
	Q ~ IGE ~ %	63	61	62	61	62	60	61	59

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3673_3_CL
SAF

Figure 9. Hover Torque Required (Sheet 9 of 15)

701C

701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	105	105	105	105	106	106	106	106
	Q ~ IGE ~ %	89	89	90	90	90	90	91	91
1000	GW ~ 100 LB	220	220	220	220	220	220	220	220
	Q ~ OGE ~ %	106	106	107	107	108	108	109	109
	Q ~ IGE ~ %	90	90	91	91	92	92	92	92
2000	GW ~ 100 LB	220	220	220	220	220	220	220	218
	Q ~ OGE ~ %	108	108	109	109	110	110	111	109
	Q ~ IGE ~ %	92	92	92	92	93	93	94	92
3000	GW ~ 100 LB	220	220	220	218	220	214	220	210
	Q ~ OGE ~ %	111	111	112	110	113	108	114	105
	Q ~ IGE ~ %	93	93	94	93	95	91	95	89
3500	GW ~ 100 LB	220	218	220	214	219	210	216	206
	Q ~ OGE ~ %	112	111	113	108	113	106	112	103
	Q ~ IGE ~ %	94	93	95	91	95	89	94	87
4000	GW ~ 100 LB	220	214	217	210	215	206	212	202
	Q ~ OGE ~ %	114	109	113	106	111	104	110	101
	Q ~ IGE ~ %	95	91	94	89	93	88	92	86
4500	GW ~ 100 LB	216	210	213	206	211	202	208	198
	Q ~ OGE ~ %	112	107	110	104	109	102	107	99
	Q ~ IGE ~ %	93	89	92	88	91	86	90	84
5000	GW ~ 100 LB	212	206	209	202	207	199	204	194
	Q ~ OGE ~ %	110	105	108	102	107	100	105	97
	Q ~ IGE ~ %	92	88	90	86	89	84	88	82
5500	GW ~ 100 LB	208	203	205	199	203	195	200	190
	Q ~ OGE ~ %	108	103	106	100	105	98	103	95
	Q ~ IGE ~ %	90	86	89	84	88	83	87	81
6000	GW ~ 100 LB	204	199	202	195	199	191	196	187
	Q ~ OGE ~ %	106	101	104	98	103	96	101	93
	Q ~ IGE ~ %	88	85	87	83	86	81	85	79

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3674_1_CL
SAF

Figure 9. Hover Torque Required (Sheet 10 of 15)

701C

701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
6500	GW ~ 100 LB	201	195	198	191	195	187	192	183
	Q ~ OGE ~ %	104	99	102	96	101	94	99	91
	Q ~ IGE ~ %	86	83	85	81	84	79	83	78
7000	GW ~ 100 LB	197	191	194	187	191	184	189	180
	Q ~ OGE ~ %	102	97	100	95	99	92	97	90
	Q ~ IGE ~ %	85	81	84	80	83	78	82	76
7500	GW ~ 100 LB	193	187	190	184	188	180	185	176
	Q ~ OGE ~ %	100	95	98	93	97	90	95	88
	Q ~ IGE ~ %	83	80	82	78	81	76	80	74
8000	GW ~ 100 LB	189	184	186	180	184	176	181	173
	Q ~ OGE ~ %	98	93	96	91	95	88	93	86
	Q ~ IGE ~ %	81	78	80	76	79	75	78	73
8500	GW ~ 100 LB	185	180	183	177	180	173	178	169
	Q ~ OGE ~ %	96	91	94	89	93	87	91	84
	Q ~ IGE ~ %	80	77	79	75	78	73	77	71
9000	GW ~ 100 LB	182	177	179	173	177	170	174	166
	Q ~ OGE ~ %	94	89	92	87	91	85	89	83
	Q ~ IGE ~ %	78	75	77	73	76	72	75	70
9500	GW ~ 100 LB	178	173	176	170	173	166	170	162
	Q ~ OGE ~ %	92	88	90	85	89	83	88	81
	Q ~ IGE ~ %	77	73	76	72	75	70	74	69
10,000	GW ~ 100 LB	175	170	172	166	170	163	167	159
	Q ~ OGE ~ %	90	86	89	84	87	81	86	79
	Q ~ IGE ~ %	75	72	74	70	73	69	72	67
10,500	GW ~ 100 LB	171	166	169	163	166	159	164	156
	Q ~ OGE ~ %	88	84	87	82	85	80	84	77
	Q ~ IGE ~ %	74	71	73	69	72	67	71	66
11,000	GW ~ 100 LB	168	163	165	160	163	156	160	153
	Q ~ OGE ~ %	87	82	85	80	84	78	82	76
	Q ~ IGE ~ %	72	69	71	68	70	66	69	64

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3674_2_CL
SAF

Figure 9. Hover Torque Required (Sheet 11 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		10		15		20		25	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
11,500	GW ~ 100 LB	164	160	162	157	160	153	157	149
	Q ~ OGE ~ %	85	81	83	79	82	76	80	74
	Q ~ IGE ~ %	71	68	70	66	69	65	68	63
12,000	GW ~ 100 LB	161	157	159	153	156	150	154	146
	Q ~ OGE ~ %	83	79	82	77	80	75	79	73
	Q ~ IGE ~ %	69	66	68	65	67	63	66	62
12,500	GW ~ 100 LB	158	153	155	150	153	147	151	143
	Q ~ OGE ~ %	81	77	80	75	79	73	77	71
	Q ~ IGE ~ %	68	65	67	64	66	62	65	60
13,000	GW ~ 100 LB	155	150	152	147	150	144	147	140
	Q ~ OGE ~ %	80	76	78	74	77	72	75	69
	Q ~ IGE ~ %	66	64	65	62	65	61	63	59
13,500	GW ~ 100 LB	152	147	149	144	147	141	144	137
	Q ~ OGE ~ %	78	74	77	72	75	70	74	68
	Q ~ IGE ~ %	65	62	64	61	63	59	62	58
14,000	GW ~ 100 LB	149	144	146	141	144	138	141	134
	Q ~ OGE ~ %	76	73	75	71	74	69	72	66
	Q ~ IGE ~ %	64	61	63	60	62	58	61	56
14,500	GW ~ 100 LB	146	141	143	138	141	135	138	132
	Q ~ OGE ~ %	75	71	73	69	72	67	70	65
	Q ~ IGE ~ %	62	60	61	58	61	57	59	55
15,000	GW ~ 100 LB	143	138	140	135	138	132		
	Q ~ OGE ~ %	73	70	72	68	71	66		
	Q ~ IGE ~ %	61	59	60	57	59	56		
15,500	GW ~ 100 LB	140	136	137	133	135	130	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	72	68	70	66	69	64		
	Q ~ IGE ~ %	60	57	59	56	58	55		
16,000	GW ~ 100 LB	137	133	134	130	132	127		
	Q ~ OGE ~ %	70	67	69	65	68	63		
	Q ~ IGE ~ %	59	56	58	55	57	53		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3674_3_CL
SAF

Figure 9. Hover Torque Required (Sheet 12 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)

HP ~FT	ATF *	FREE AIR TEMPERATURE ~°C							
		30		35		40		45	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
0	GW ~ 100 LB	220	220	220	220	220	216	220	210
	Q ~ OGE ~ %	107	107	108	108	109	106	110	102
	Q ~ IGE ~ %	91	91	92	92	93	90	93	87
500	GW ~ 100 LB	220	220	220	218	220	212	220	206
	Q ~ OGE ~ %	108	108	109	107	110	104	111	100
	Q ~ IGE ~ %	92	92	93	91	94	89	94	86
1000	GW ~ 100 LB	220	220	220	214	220	208	216	202
	Q ~ OGE ~ %	109	109	110	105	111	102	109	98
	Q ~ IGE ~ %	93	93	94	90	94	87	93	84
1500	GW ~ 100 LB	220	216	220	210	218	204	212	199
	Q ~ OGE ~ %	111	108	112	103	111	100	107	96
	Q ~ IGE ~ %	94	91	94	88	94	85	91	83
2000	GW ~ 100 LB	220	212	219	206	214	200	208	195
	Q ~ OGE ~ %	112	105	113	101	109	98	105	95
	Q ~ IGE ~ %	94	90	95	86	92	84	89	81
2500	GW ~ 100 LB	220	208	215	202	210	197	204	191
	Q ~ OGE ~ %	113	104	110	99	107	96	103	93
	Q ~ IGE ~ %	95	88	93	85	90	82	87	79
3000	GW ~ 100 LB	216	204	211	198	206	193	200	187
	Q ~ OGE ~ %	112	102	108	97	105	94	101	91
	Q ~ IGE ~ %	94	86	91	83	89	81	86	78
3500	GW ~ 100 LB	212	200	207	194	202	189	197	184
	Q ~ OGE ~ %	110	100	106	96	103	92	99	89
	Q ~ IGE ~ %	92	85	89	81	87	79	84	76
4000	GW ~ 100 LB	208	197	203	191	198	185	193	180
	Q ~ OGE ~ %	107	98	104	94	100	90	97	87
	Q ~ IGE ~ %	90	83	88	80	85	77	82	75
4500	GW ~ 100 LB	204	193	199	187	194	182	189	177
	Q ~ OGE ~ %	105	96	102	92	99	89	95	85
	Q ~ IGE ~ %	88	81	86	78	83	76	81	73
5000	GW ~ 100 LB	200	189	196	183	190	178	185	173
	Q ~ OGE ~ %	103	94	100	90	97	87	93	84
	Q ~ IGE ~ %	87	80	84	77	82	74	79	72
5500	GW ~ 100 LB	196	185	192	180	187	175	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	101	92	98	88	95	85		
	Q ~ IGE ~ %	85	78	83	75	80	73		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR 10 MINUTE ENGINE Q LIMIT.

AB3675_1_CL
SAF

Figure 9. Hover Torque Required (Sheet 13 of 15)

701C

701D/CC

**MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)**

HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C							
		30		35		40		45	
		1.0	0.9	1.0	0.9	1.0	0.9	1.0	0.9
6000	GW ~ 100 LB	192	182	188	176	183	171	178	166
	Q ~ OGE ~ %	99	90	96	86	93	83	89	80
	Q ~ IGE ~ %	83	77	81	74	78	71	76	69
6500	GW ~ 100 LB	189	178	184	173	179	168	174	163
	Q ~ OGE ~ %	97	88	94	85	91	82	87	79
	Q ~ IGE ~ %	82	75	79	72	77	70	74	67
7000	GW ~ 100 LB	185	174	181	169	176	164		
	Q ~ OGE ~ %	95	86	92	83	89	80		
	Q ~ IGE ~ %	80	74	78	71	75	68		
7500	GW ~ 100 LB	181	171	177	166	172	161		
	Q ~ OGE ~ %	93	85	90	81	87	78		
	Q ~ IGE ~ %	78	72	76	69	74	67		
8000	GW ~ 100 LB	178	168	173	162	169	158		
	Q ~ OGE ~ %	91	83	88	79	85	76		
	Q ~ IGE ~ %	77	70	74	68	72	65		
8500	GW ~ 100 LB	174	164	170	159	165	154		
	Q ~ OGE ~ %	89	81	86	78	83	75		
	Q ~ IGE ~ %	75	69	73	66	71	64		
9000	GW ~ 100 LB	171	161	166	156				
	Q ~ OGE ~ %	87	79	84	76	ABOVE ENGINE AMBIENT TEMP LIMIT			
	Q ~ IGE ~ %	73	68	71	65				
9500	GW ~ 100 LB	167	158	163	152				
	Q ~ OGE ~ %	85	78	83	74				
	Q ~ IGE ~ %	72	66	70	63				
10,000	GW ~ 100 LB	163	154	159	149				
	Q ~ OGE ~ %	83	76	81	73				
	Q ~ IGE ~ %	70	65	68	62				
10,500	GW ~ 100 LB	160	151	156	146				
	Q ~ OGE ~ %	82	74	79	71				
	Q ~ IGE ~ %	69	63	67	61				
11,000	GW ~ 100 LB	157	148	153	143				
	Q ~ OGE ~ %	80	73	77	69				
	Q ~ IGE ~ %	67	62	65	59				
11,500	GW ~ 100 LB	153	145	150	140				
	Q ~ OGE ~ %	78	71	75	68				
	Q ~ IGE ~ %	66	61	64	58				

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR 10 MINUTE ENGINE Q LIMIT.

AB3675_2_CL
SAF

Figure 9. Hover Torque Required (Sheet 14 of 15)

701C 701D/CC

MAXIMUM OGE HOVER WEIGHT AND TORQUE REQUIRED
ANTI-ICE OFF T700-GE-701C AND T700-GE-701D/CC ENGINE
(CONTINUED)



HP ~FT	ATF *	FREE AIR TEMPERATURE ~ °C			
		50		55	
		1.0	0.9	1.0	0.9
0	GW ~ 100 LB	219	204	211	197
	Q ~ OGE ~ %	109	98	104	94
	Q ~ IGE ~ %	93	84	89	81
500	GW ~ 100 LB	214	200	207	193
	Q ~ OGE ~ %	107	97	102	92
	Q ~ IGE ~ %	91	83	88	79
1000	GW ~ 100 LB	210	196	203	189
	Q ~ OGE ~ %	105	95	100	90
	Q ~ IGE ~ %	90	81	86	78
1500	GW ~ 100 LB	206	193	199	186
	Q ~ OGE ~ %	103	93	98	89
	Q ~ IGE ~ %	88	80	84	76
2000	GW ~ 100 LB	202	189	195	182
	Q ~ OGE ~ %	101	91	96	87
	Q ~ IGE ~ %	86	78	82	75
2500	GW ~ 100 LB	198	185	ABOVE ENGINE AMBIENT TEMP LIMIT	
	Q ~ OGE ~ %	99	89		
	Q ~ IGE ~ %	84	76		

* GW ~ 100 LB = GW DIVIDED BY 100 LB

Q = TORQUE %

OGE = 100 FT WHEEL HEIGHT

IGE = 10 FT WHEEL HEIGHT

MAXIMUM HOVER WEIGHT IS LIMITED TO 22,000 LB

120% Q TRANSMISSION LIMIT OR

10 MINUTE ENGINE Q LIMIT.

AB3676CL
SAF

Figure 9. Hover Torque Required (Sheet 15 of 15)

701C

701D/CC

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% - % RPM R = 100%) PRESSURE ALTITUDE - FEET					
	-1000	-500	0	500	1000	1500
55	736	740	744	748	753	758
50	721	724	728	733	737	742
45	706	709	713	717	721	725
39	690	692	696	700	704	707
37	684	687	690	694	698	702
35	679	681	685	689	692	695
33	673	676	679	683	687	690
31	668	670	674	677	681	685
29	662	665	668	671	675	679
27	657	659	662	666	670	673
25	651	654	657	660	664	667
23	645	648	651	655	658	662
21	639	642	645	649	652	656
19	634	636	640	643	647	650
17	628	631	634	638	641	644
15	623	625	629	632	635	638
13	617	620	623	626	630	633
11	612	614	618	621	624	627
9	606	609	612	616	619	622
7	600	603	607	610	613	616
5	595	598	601	605	608	611
3	589	592	595	599	602	605
1	584	586	590	593	597	600
-1	577	579	583	586	590	593
-3	571	574	577	581	584	587
-5	566	568	572	575	578	581
-7	560	563	566	569	573	576
-9	554	557	560	564	567	570
-11	549	552	555	558	561	564
-13	543	546	549	553	556	559
-15	538	540	544	547	550	553
-17	532	535	538	541	544	547
-19	526	529	532	536	539	542
-21	521	523	527	530	533	536
-23	515	518	521	524	527	530
-25	510	512	515	519	522	524
-27	504	507	510	513	516	519
-29	498	501	504	507	510	513
-31	492	495	498	501	504	507
-33	487	490	493	496	499	501
-35	482	484	487	490	493	496
-37	476	478	481	484	487	490
-39	470	473	476	479	482	484
-45	453	456	459	462	465	467
-50	439	441	444	447	450	453
-55	425	427	430	433	436	438

AB2790_1A_CL
SA

Figure 10. TGT Reference Table (Sheet 1 of
3) **701C** **701D/CC**

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% - % RPM R = 100%) PRESSURE ALTITUDE - FEET					
	2000	2500	3000	3500	4000	4500
55	763	769	775	781	787	792
50	747	752	758	764	770	776
45	731	736	741	747	753	759
39	712	716	721	725	732	738
37	706	710	714	720	725	731
35	700	704	709	713	719	724
33	694	698	703	707	712	717
31	689	692	697	702	707	711
29	683	687	691	696	701	705
27	677	681	685	690	695	700
25	671	675	680	684	689	694
23	666	669	674	678	683	688
21	660	664	668	672	677	682
19	654	658	662	667	671	676
17	648	652	656	661	665	670
15	642	646	650	655	659	664
13	636	640	644	649	654	658
11	631	634	638	643	647	652
9	625	629	633	637	641	646
7	620	623	627	631	635	640
5	614	618	621	625	630	634
3	609	612	616	620	624	629
1	603	607	610	614	618	622
-1	596	600	603	607	611	615
-3	591	594	598	602	606	609
-5	585	589	592	596	600	604
-7	579	583	587	590	594	598
-9	574	577	581	585	589	592
-11	568	571	575	579	583	586
-13	562	566	569	573	577	581
-15	556	560	564	567	571	575
-17	551	555	558	562	565	569
-19	545	549	552	556	559	563
-21	539	543	548	550	554	557
-23	534	537	540	544	548	551
-25	528	531	535	538	542	545
-27	522	526	529	532	536	539
-29	516	520	523	525	530	534
-31	511	514	517	521	524	528
-33	505	508	511	515	518	522
-35	499	502	506	509	512	516
-37	493	497	500	503	506	510
-39	487	491	494	497	500	504
-45	470	473	476	479	483	486
-50	456	459	462	465	468	471
-55	441	444	447	450	454	457

AB2790_2CL

SA

Figure 10. TGT Reference Table (Sheet 2 of 3)

701C

701D/CC

FAT °C	ENGINE HEALTH INDICATOR TEST (HIT) TGT REFERENCE TABLE (TRQ = 60% - % RPM R = 100%) PRESSURE ALTITUDE - FEET						
	5000	5500	6000	7000	8000	9000	10000
55	796	801	805	814	825	835	846
50	782	788	792	802	811	822	832
45	764	771	776	788	798	808	818
39	744	750	755	768	780	792	802
37	737	743	748	761	773	787	796
35	730	736	741	754	766	779	791
33	723	729	735	746	759	772	786
31	716	722	727	739	752	765	779
29	710	715	720	732	745	758	771
27	704	709	714	725	737	750	764
25	698	703	707	718	730	743	756
23	692	697	701	711	723	736	749
21	686	691	695	705	715	728	742
19	680	685	689	699	709	721	734
17	674	679	683	693	703	714	727
15	668	673	677	686	697	708	719
13	662	666	671	680	690	701	712
11	656	660	665	674	684	695	706
9	650	654	659	668	678	689	699
7	644	648	652	662	671	682	693
5	637	642	646	656	665	676	687
3	631	636	640	649	659	669	680
1	626	629	634	643	653	663	674
-1	618	622	625	635	645	655	666
-3	613	617	620	629	639	649	659
-5	607	611	615	623	632	642	652
-7	601	605	609	617	625	636	646
-9	596	599	603	611	620	630	639
-11	590	594	597	606	614	623	633
-13	584	588	591	600	608	617	627
-15	578	582	586	594	602	611	620
-17	572	576	580	588	596	605	614
-19	566	570	574	582	591	599	608
-21	560	564	568	577	585	593	602
-23	555	558	562	571	579	587	596
-25	549	552	556	565	573	581	589
-27	543	547	550	559	567	575	583
-29	537	541	545	553	561	569	577
-31	531	535	539	547	554	563	571
-33	525	529	533	541	548	556	565
-35	519	523	527	539	542	550	558
-37	513	517	521	529	536	544	552
-39	507	511	515	523	530	538	546
-45	489	493	497	505	512	519	527
-50	475	478	482	490	497	504	512
-55	460	464	467	475	481	489	496

AB2790_3CL

SA

Figure 10. TGT Reference Table (Sheet 3 of
3) **701C** **701D/CC**

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is fluid and cursive, with the first name "Joyce" and last name "Morrow" clearly legible.

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0609507

DISTRIBUTION:

To be distributed in accordance with Initial Distribution
Number (IDN) 311310, requirements for TM 1-1520-237-
CL.

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
1 decimeter = 10 centimeters = 3.94 inches
1 meter = 10 decimeters = 39.37 inches
1 dekameter = 10 meters = 32.8 feet
1 hectometer = 10 dekameters = 328.08 feet
1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
1 decigram = 10 centigrams = 1.54 grains
1 gram = 10 decigrams = .035 ounce
1 dekagram = 10 grams = .35 ounce
1 hectogram = 10 dekagrams = 3.52 ounces
1 kilogram = 10 hectograms = 2.2 pounds
1 quintal = 100 kilograms = 220.46 pounds
1 metric ton = 10 quintals = 1.1 short ton

Liquid Measure

1 centiliter = 10 milliliters = .34 fl. ounce
1 deciliter = 10 centiliters = 3.38 fl. ounces
1 liter = 10 deciliters = 33.81 fl. ounces
1 dekaliter = 10 liters = 2.64 gallons
1 hectoliter = 10 dekaliters = 26.42 gallons
1 kiloliter = 10 hectoliters = 264.18 gallons

Temperature Conversion

$$^{\circ}\text{C to }^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$$

$$^{\circ}\text{F to }^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$

PIN: 073160-000